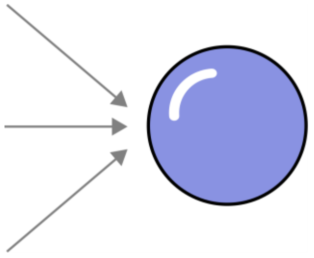
Introduction to 3D Modeling

Yogesh H Kulkarni



Introduction

Introduction

- How to model real world objects? - Design
- How to put forth ideas in visual manner – Communication
- How to verify that design serves the purpose – Analysis
- How to get it made? – Manufacturing
- All of the above can happen without Computers. But
- Better if assisted by Computers/Software

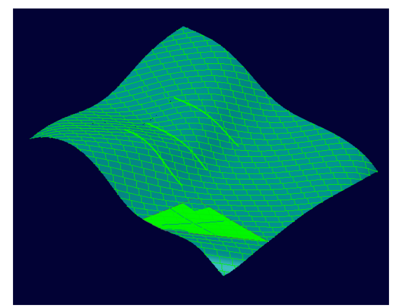
- That’s why : Computer Aided < > (CAx)

History

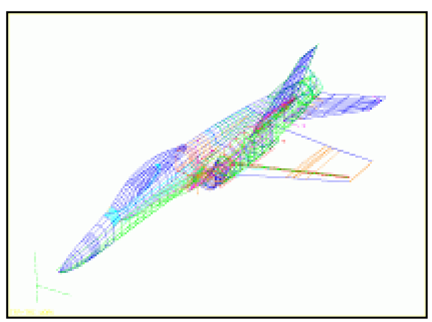
- The first source of CAD resulted from attempts to automate the drafting process.
- These developments were pioneered by the General Motors Research Laboratories in the early 1960s.
- CAD became more widely used after 1970 because of technological advancements.
- CAD allowed users to design products much quicker without the production of an actual product.

Evolution of CAD Technology

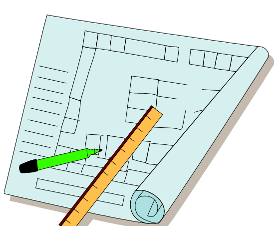
Solid model



Surface model



Wire frame model

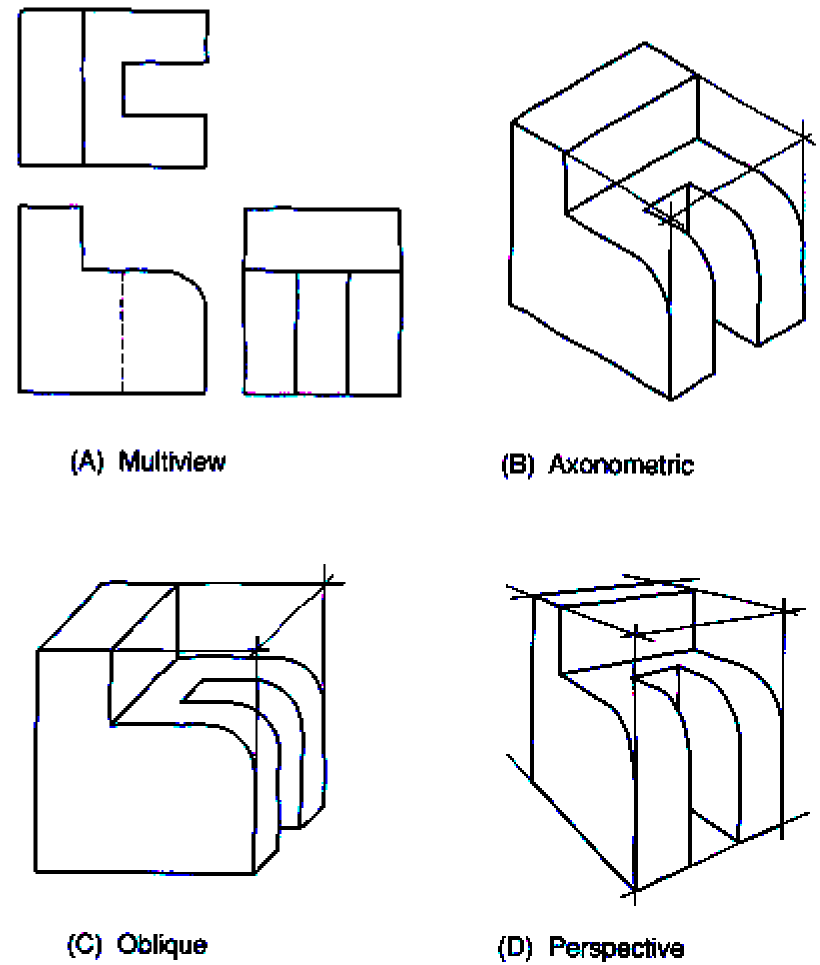


Drawing

Manual drafting
Since 1970's: electronic drafting board

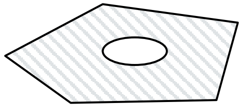
Manual Drafting

- 2D representations used to represent 3D objects
  - multi-view drawings
  - pictorials
- Standards and conventions developed so that 3D object could be built from drawings
- Drawings created manually or using 2D CAD
- Difficult to visualize, error-prone, time-consuming

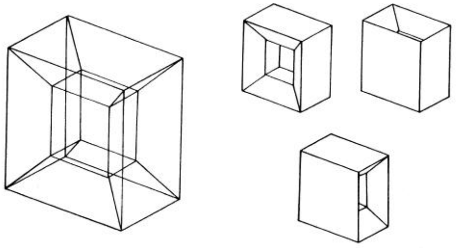


CAD - Types

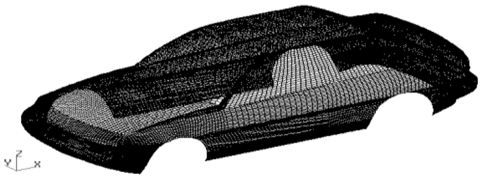
- 2D model: Point, line, circular arc, planar curve



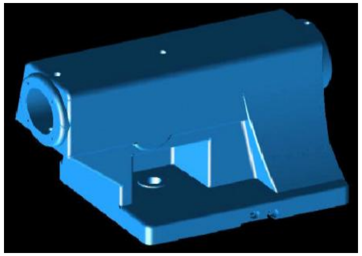
- 3D model
  - Wire frame



  - Surface



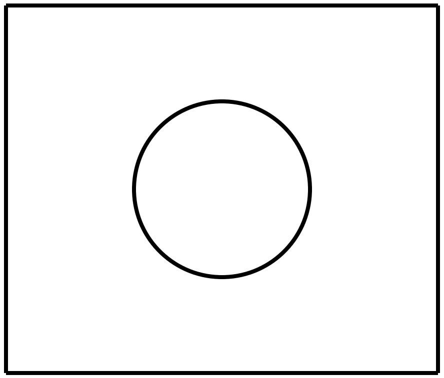
  - Solid

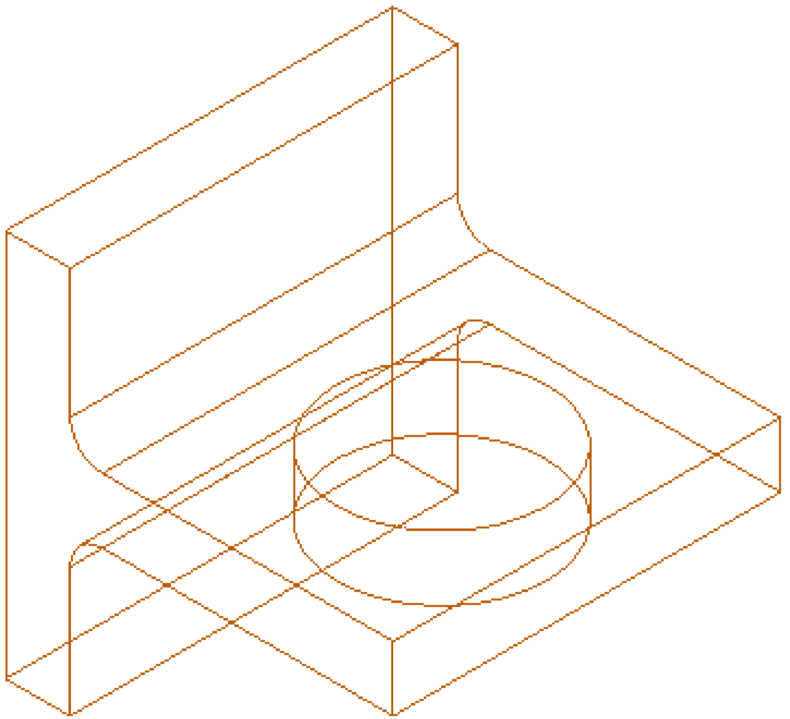


Advantages and Disadvantages of each?

2D CAD

- Simply replaces manual drawing
- Provides a set of drawing tools to create 2D elements
  - Lines, circles, arcs, etc.
- More accurate, easier changes to drawings
- Still no 3D representation of the object
- Example: AutoCAD

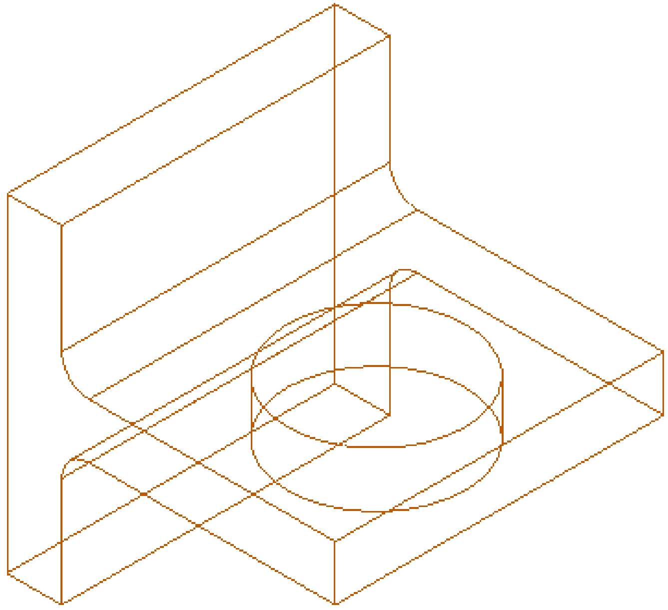




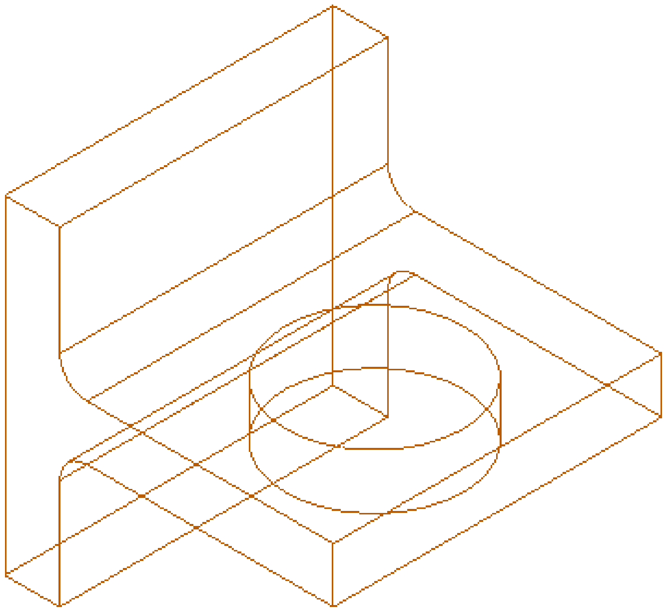
Early 1980's: wire frame geometry

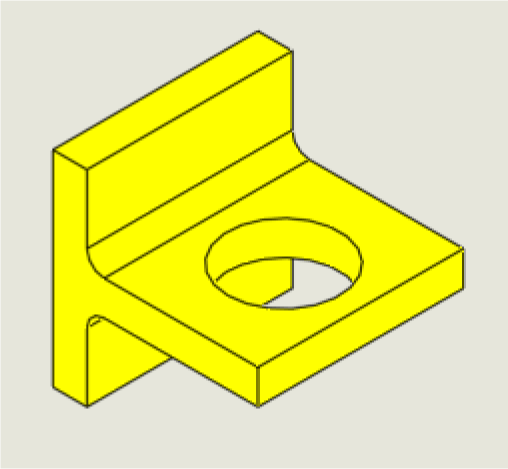
3D Wire frame Modeling

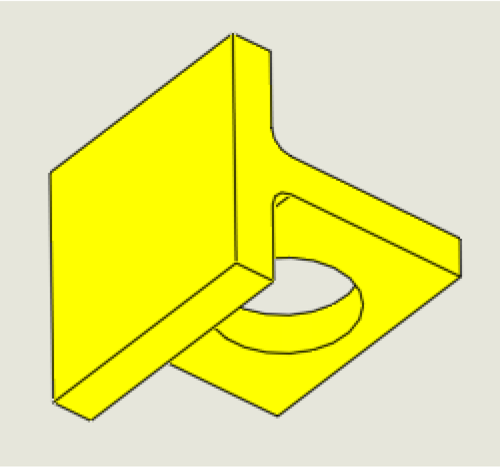
- Geometric entities are lines and curves in 3D
- Volume or surfaces of object not defined
- Easy to store and display
- Hard to interpret - ambiguous



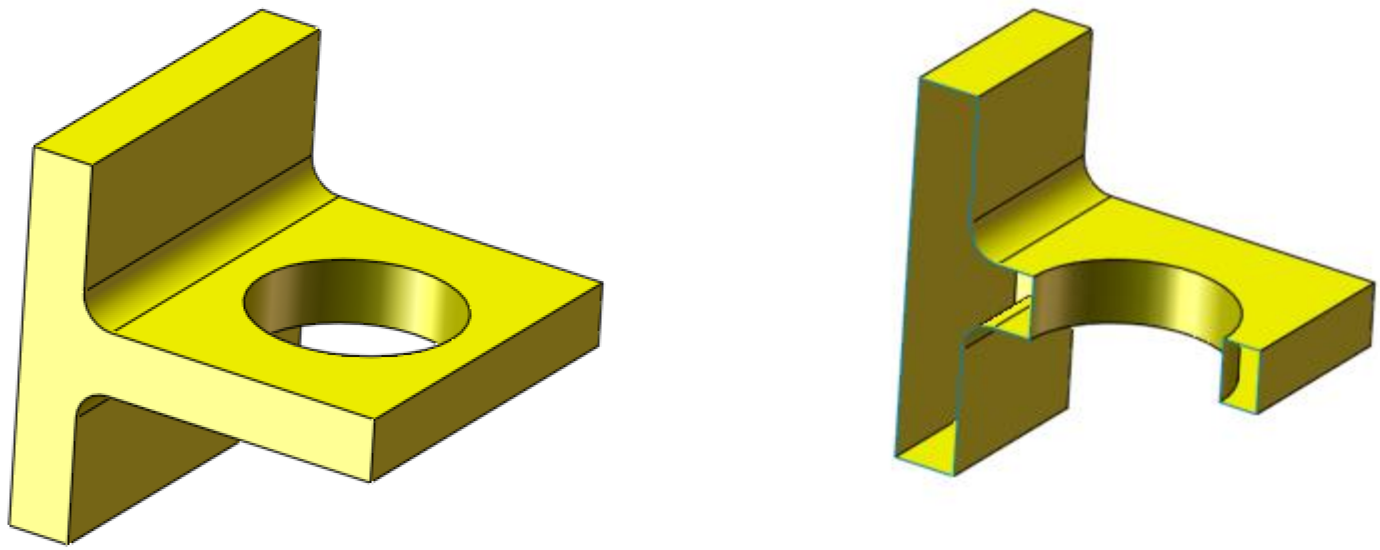
What is this?







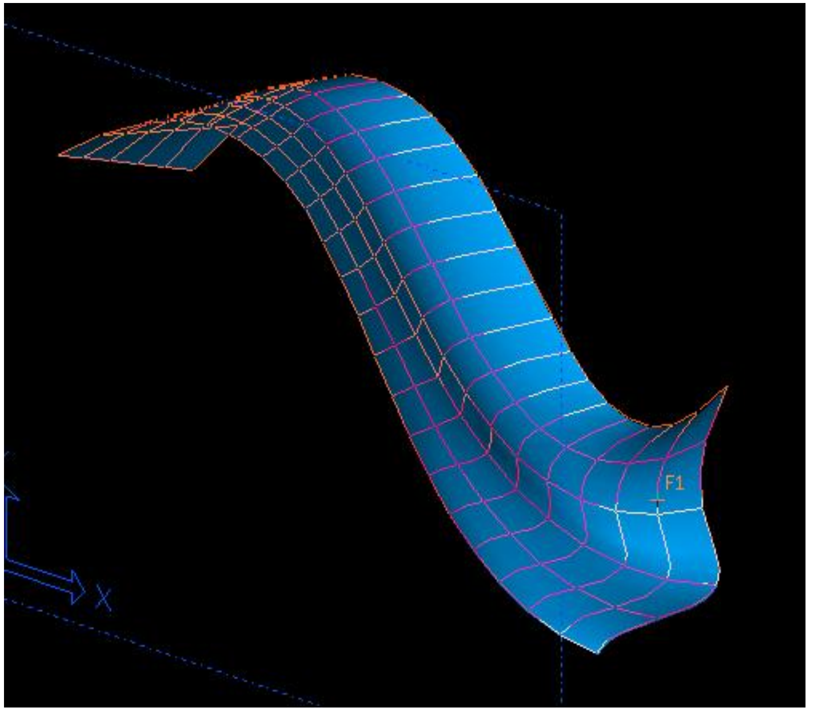
Problems with wire frame models



Late 1980’s: Surface Modeling

3D Surface Modeling

- Models 2D surfaces in 3D space
- All points on surface are defined
  - useful for machining, visualization, etc.
- Surfaces have no thickness, objects have no volume or solid properties
- Surfaces may be open



A Surface Model created using Alias StudioTools

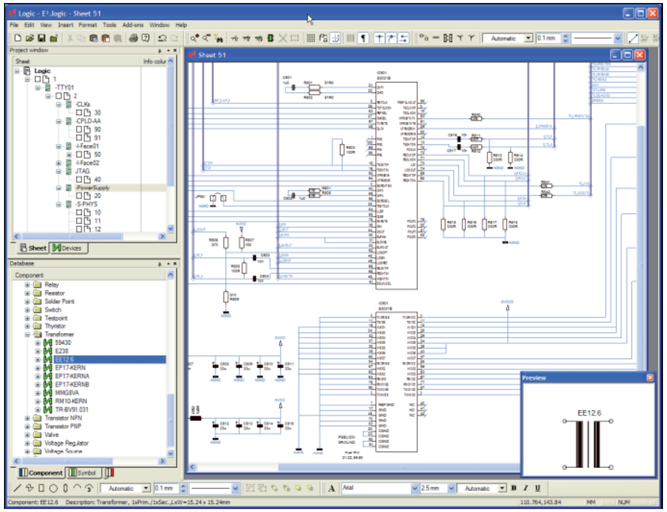
Surface Model created using Rhino

Why draw 3D Models?

- 3D models are easier to interpret.
- Less expensive than building a physical model.
- 3D models can be altered easily, create more concepts.
- 3D models can be used to perform engineering analysis, finite element analysis (stress, deflection, thermal.....) and motion analysis.
- 3D models can be used directly in manufacturing, Computer Numerical Control (CNC).

2D Applications

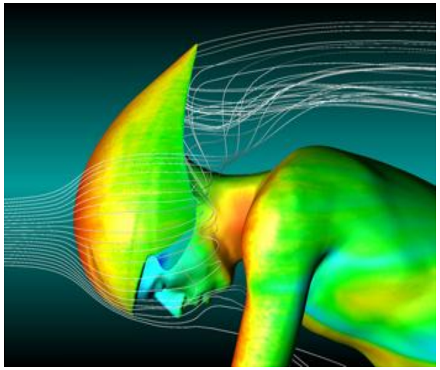
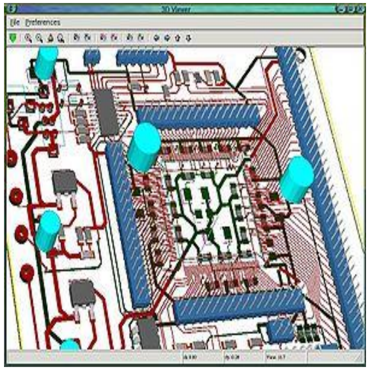
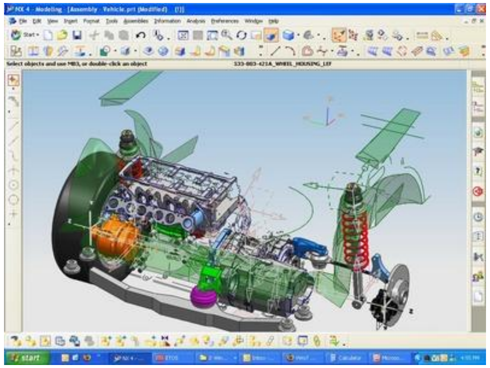
- Drafting – sketches, architectures, Drawings
- Art – Sketches, painting
- Electronic layouts, circuit design



AUTODESK.

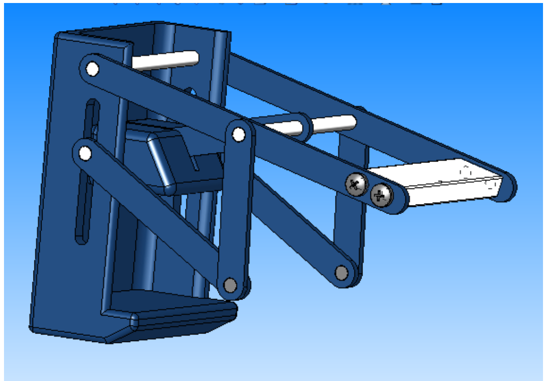
3D Applications

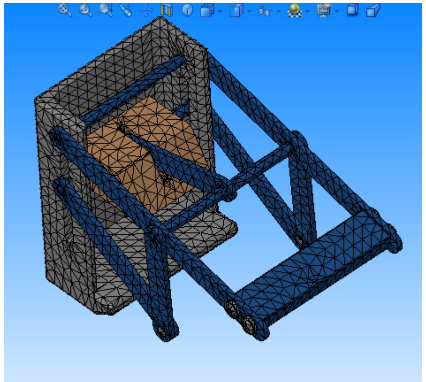
- CAD (Computer Aided Design)
- CAM (Computer Aided Manufacturing)
- CAE (Computer Aided Engineering) Finite Element Method
- CG (Computer Graphics)

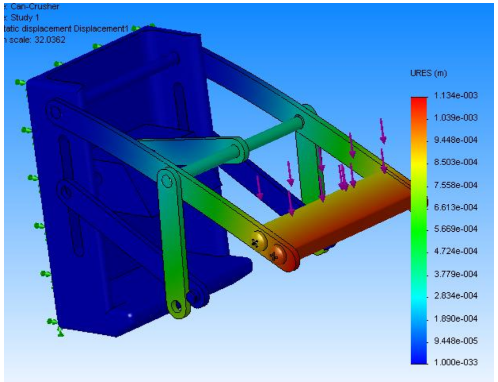




Basics of Finite Element Analysis (FEA)

- A complex problem is divided into a smaller and simpler problems that can be solved by using the existing knowledge of mechanics of materials and mathematical tools
- Modern mechanical design involves complicated shapes, sometimes made of different materials that as a whole cannot be solved by existing mathematical tools. Engineers need the FEA to evaluate their designs







Computer Numerical Control (CNC)

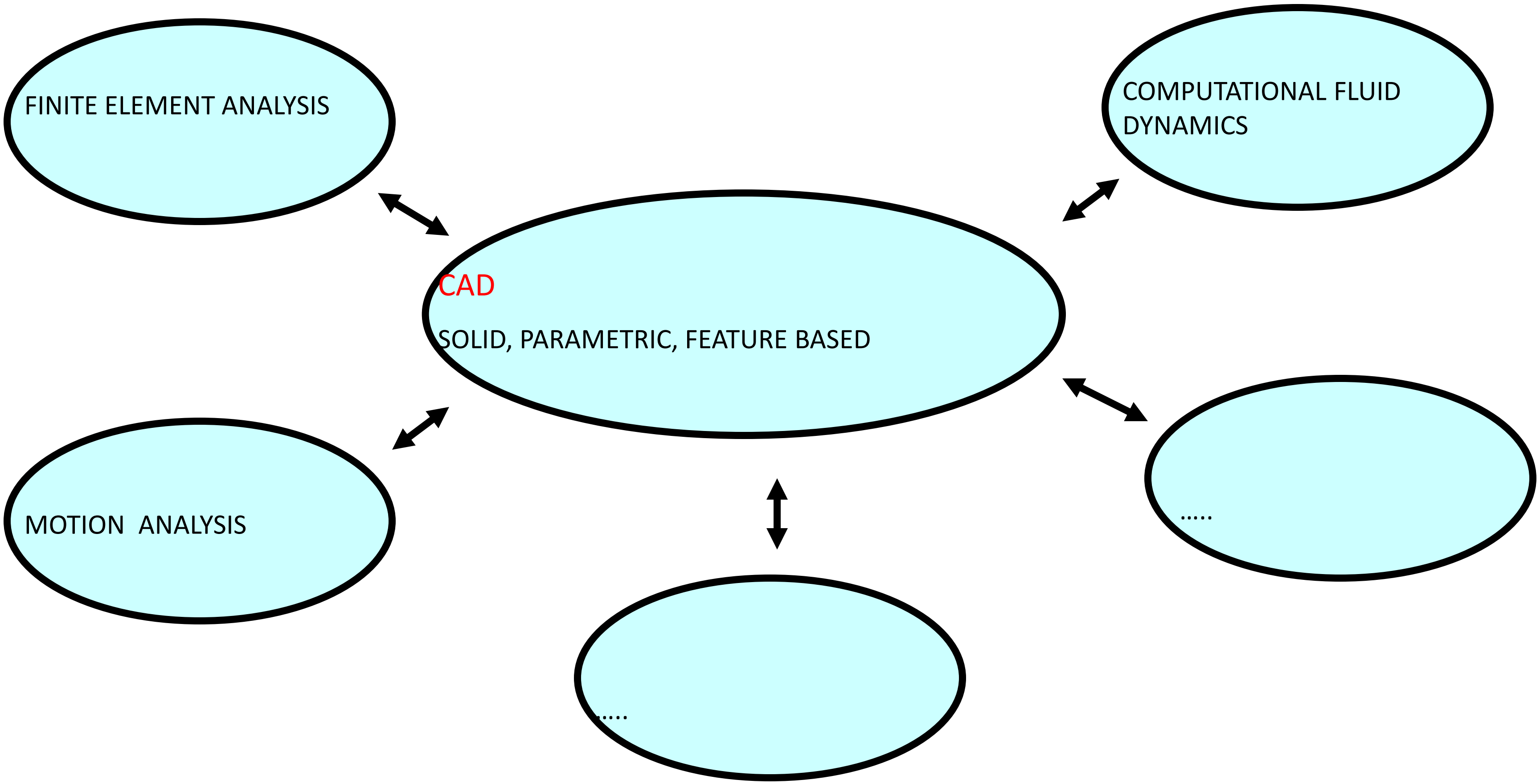
A CNC machine is an NC machine with the added feature of an on-board computer.

Modelling

Solid, parametric, feature based modeling

- Complete and unambiguous
- Solid - models have volume, and mass properties
- Feature based - geometry built up by adding and subtracting features
- Parametric - geometry can be modified by changing dimensions

MODERN CAE TOOLS



CAD (Computer Aided Design) is at the hub of other CAE (Computer Aided Engineering) tools

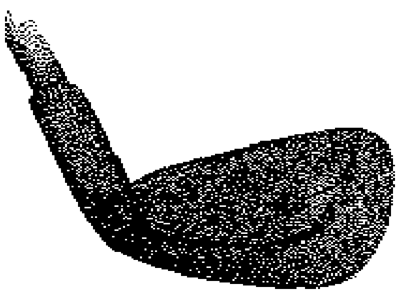
Solids

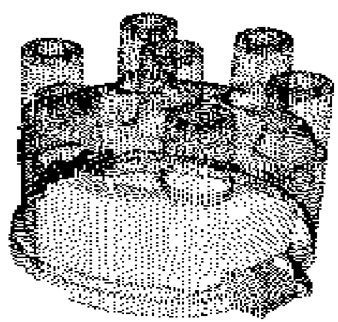
What is Solid?

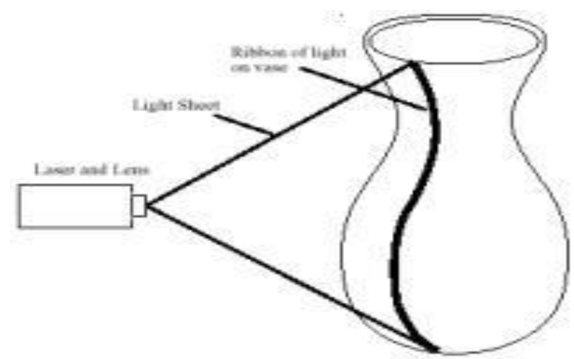
- Define Solid?
- How would you represent Solid in software (data model)?

Cloud of points

- The simplest form
- Unorganized / organized points
- Too many points to represent the desired shape
- Hard to handle → further processing is required
- Obtained by digitizing
    - CMM (coordinate measuring machine)
    - Laser range scanner
    - …

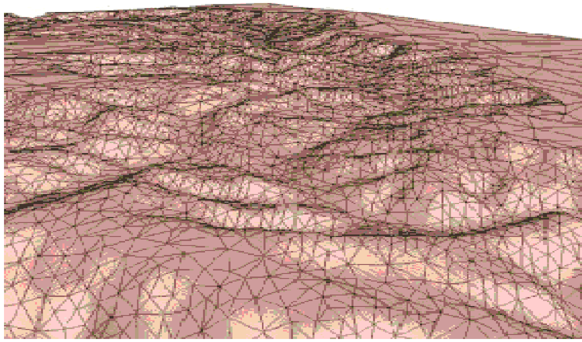


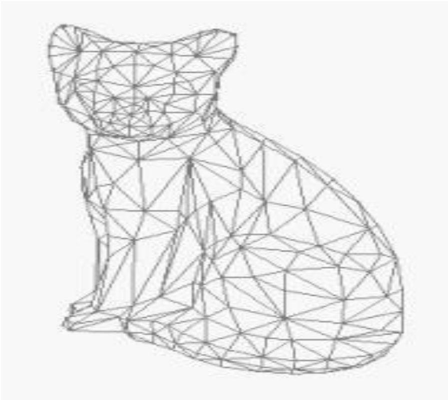




Mesh

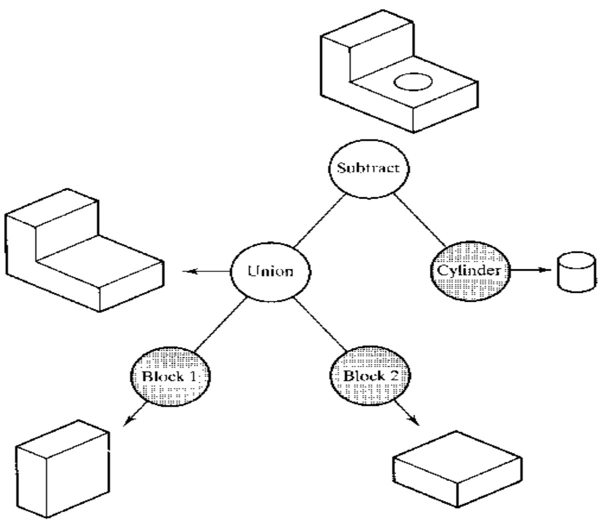
- Most popular approximation model
- Graphics, RP, CAD/CAM, DMU, CAE
- Hard to handle
- Triangular mesh, Quad mesh, General polygonal mesh
- Create mesh by
  - **triangulating** cloud of points
  - **faceting** exact surface model
- Example: 123D Catch





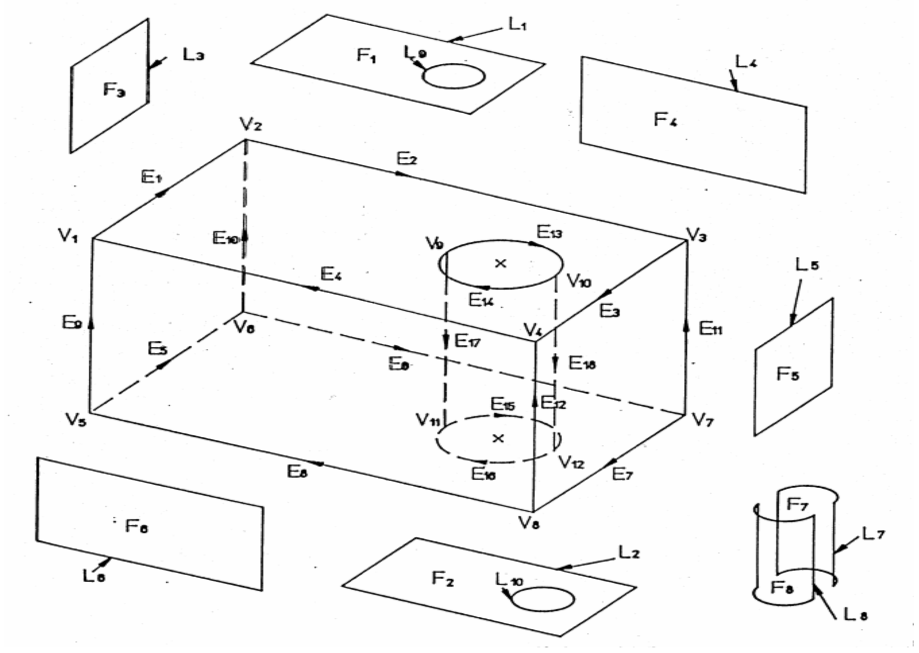
What to store : Modeling procedure

- Procedural model: CSG (Constructive Solid Geometry)
- Primitive solids with Boolean operation



What to store : result

- B-Rep (Boundary representation) model : Modeling using bounding surfaces
- Topology : connectivity
- Geometry : shape



B-Rep model

- Topological element
  - Vertex
  - Edge
  - Loop (Edge list)
  - Face
  - Lump
  - Body

- Geometrical element
  - Point
  - Curve
  - Composite curve
  - Surface, trimmed surface
  - N/A
  - N/A

Euler-Poincare formula:

For a polyhedron

$$V - E + F - 2 = 0$$

- V = Vertices
- E = Edges
- F = Faces

Example: A tetrahedron has four vertices, four faces, and six edges

$$4-6+4=2.$$

Extension of solids

A solid can have holes

A face may have a loop or ring of vertices `floating‘, i.e. unconnected by edges to the other vertices of the face

Extension of Euler-Poincare formula to 2-manifolds

$$V-E+F-H=2(C-G)$$

- V = Vertices
- E = Edges
- F = Faces
- H = Holes in faces
- C = Components (or shells)
- G = Genus (holes through solid)

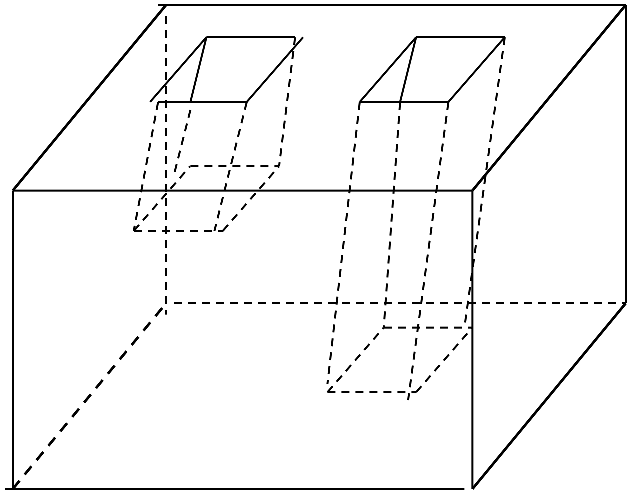
"Tweaking" (deformations, twistings, and stretchings but not tearing, or cutting) solids modifies the solid without changing the topology or the above numbers.

A solid with holes and loops

Example

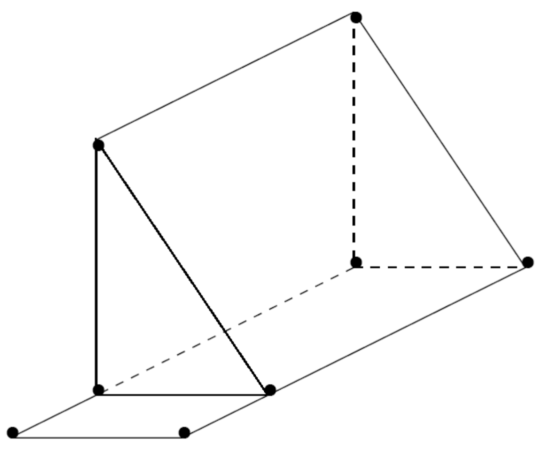
$$V - E + F - H = 2\,(C - G)$$

$$24 - 36 + 15 - 3 = 2(\,1 - 1)$$



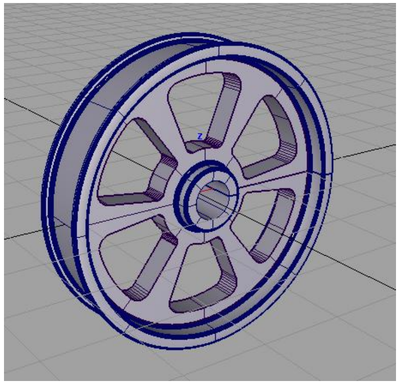
Euler Poincare' Formula

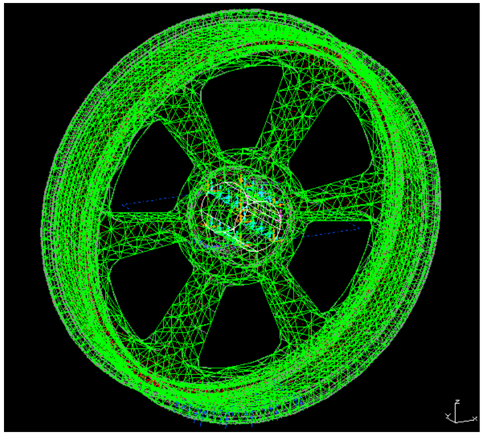
- Necessary but not sufficient condition for a valid representation.
- Example: 8 vertices, 12 edges, 6 faces



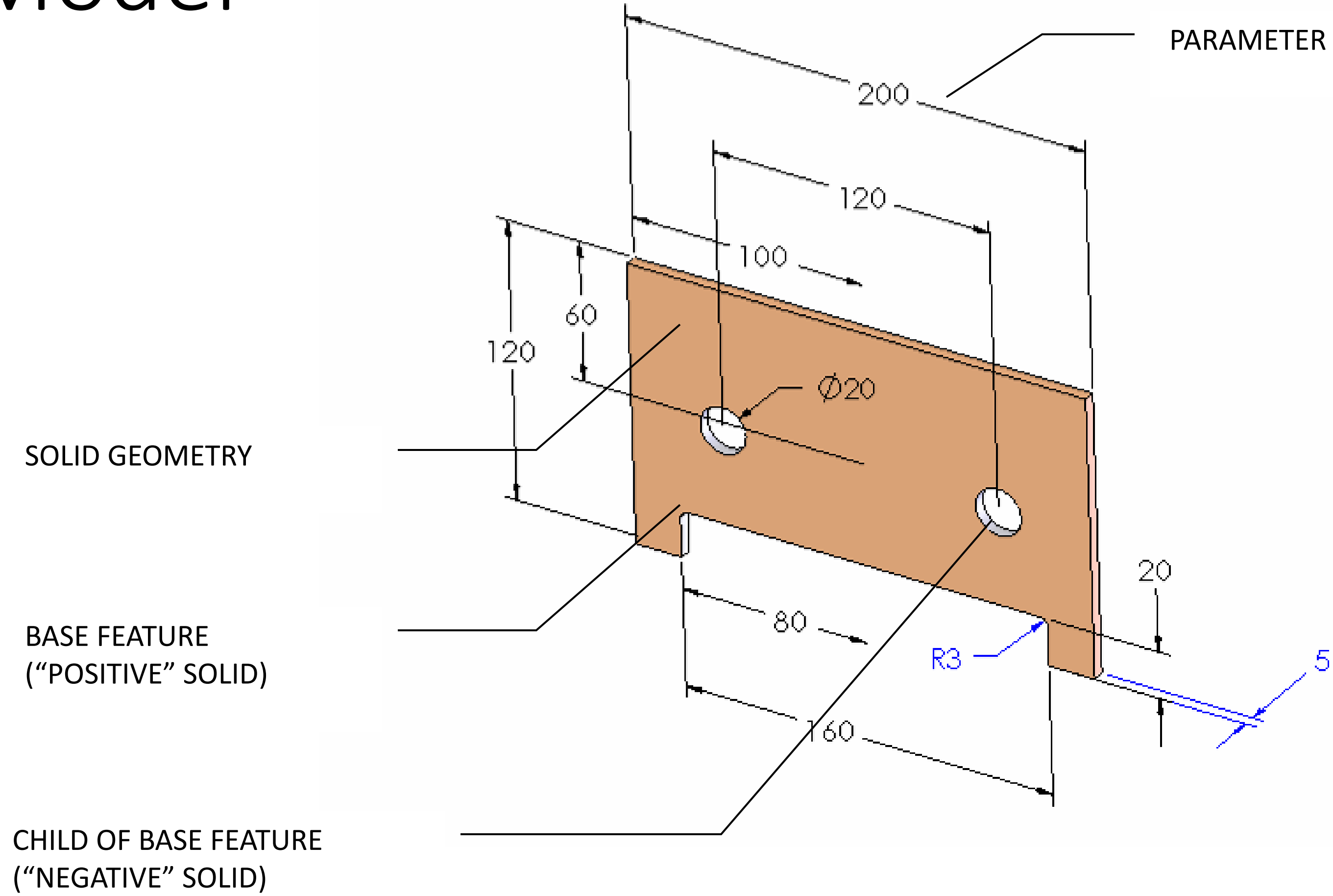
Brep vs Mesh (Design Desktop vs Catch)

- The object is represented by subdivision/discretization such as mesh and other geometric primitives.





Parametric, Feature-based Solid Model



panel 01.SLDPRT

Solid, parametric, feature-based Modeling Software

- High-end (more powerful)
  - NX (UGS)
  - Catia (Dassault Systémes)
  - Pro/Engineer (Parametric Technologies Corp.)
- Mid-Range (easier to use)
  - Solid Edge (UGS)
  - Inventor (Autodesk)
  - SolidWorks (SolidWorks Corp.)

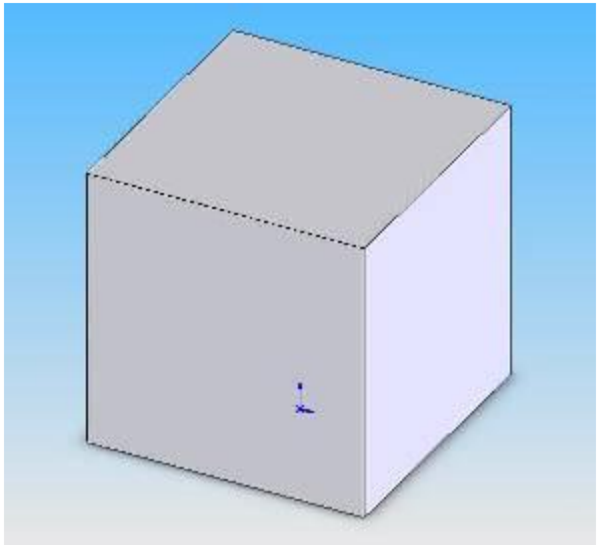
They all work basically the same way

Feature-Based Solid Modeling

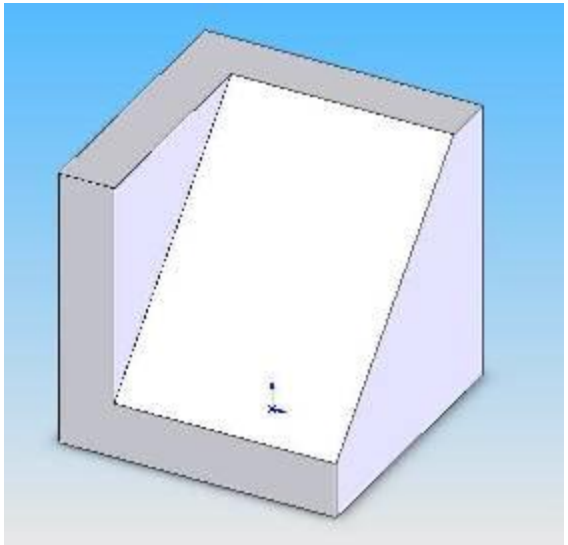
- Parts modeled by adding **features** to a base part
- Features represent “operations”
  - holes, ribs, fillets, chamfers, slots, pockets, etc.
- Material can be added or subtracted
- Features can be created by extrusion, sweeping, revolving, etc.

Feature-based Modeling Process

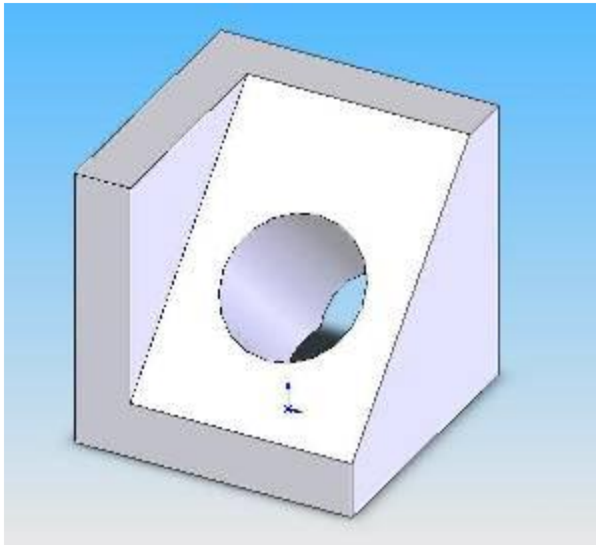
- Create base part
- Add features until final shape is achieved



Extruded Base



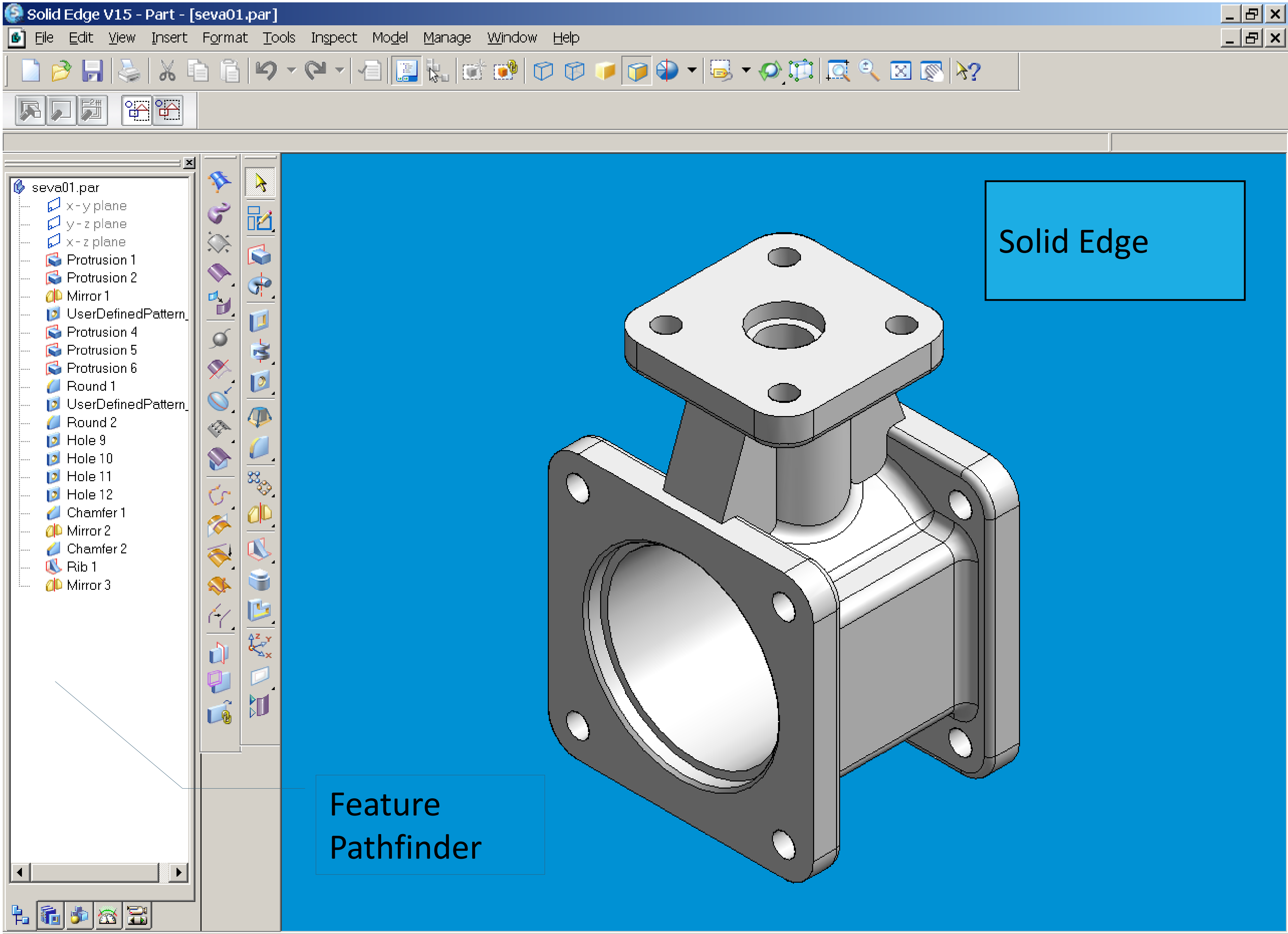
Extruded Cut



Extruded Cut

Feature History Trees

- Most feature-based modelers show the features and their order in a graphical tree view
- This view has different names, depending on the software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seva01.par
x-y plane
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Protrusion 1
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Protrusion 4
Protrusion 5
Protrusion 6
Round 1
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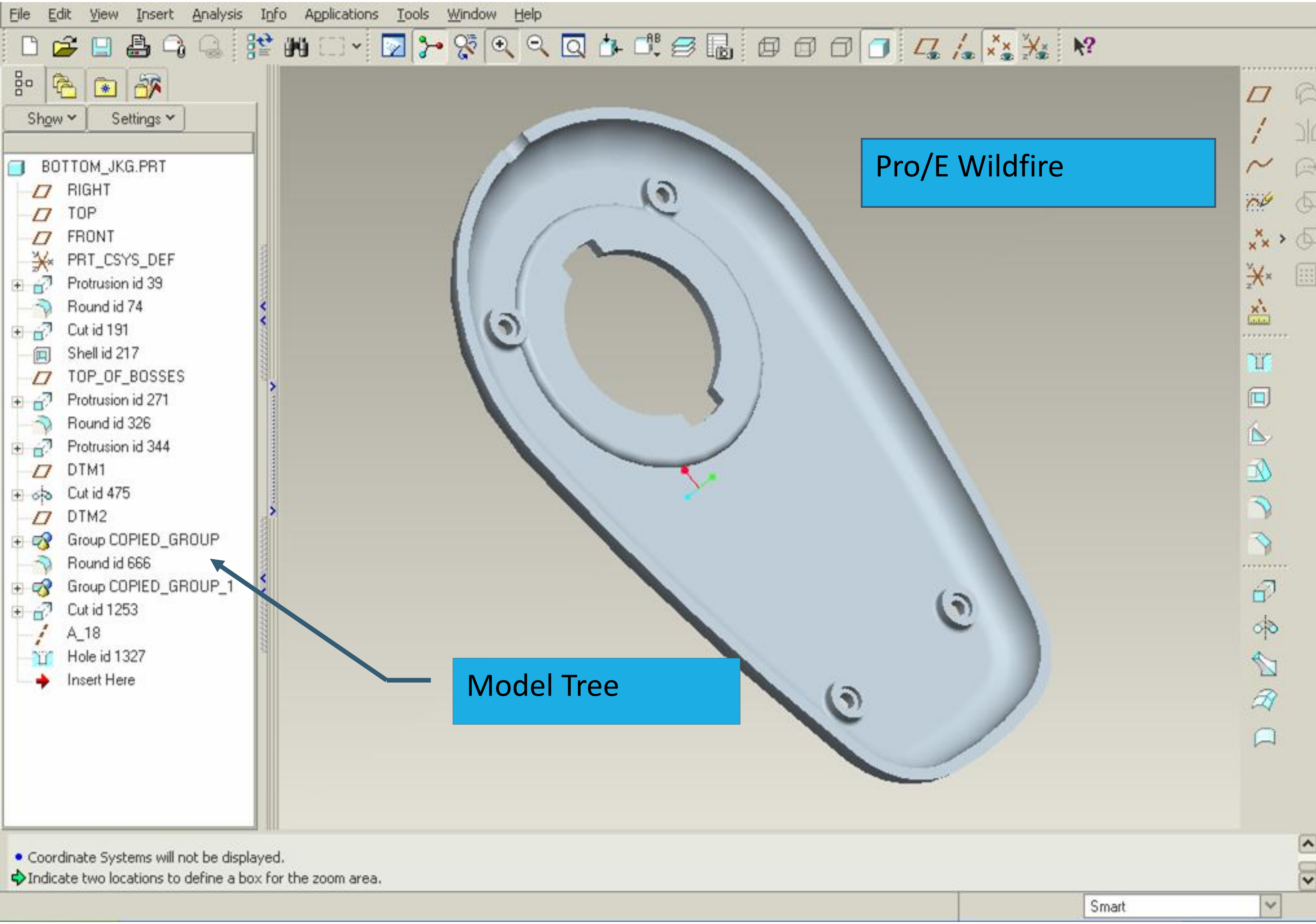
Hole 9
Hole 10
Hole 11
Hole 12
Chamfer 1
Mirror 2
Chamfer 2
Rib 1
Mirror 3
Solid Edge
Feature Pathfinder


Pro/E Wildfire
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Protrusion id 39
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Cut id 191
Shell id 217
TOP_OF_BOSSES
Protrusion id 271
Round id 326
Protrusion id 344
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Cut id 475
DTM2
Group COPIED_GROUP
Round id 666
Group COPIED_GROUP_1
Cut id 1253
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Hole id 1327
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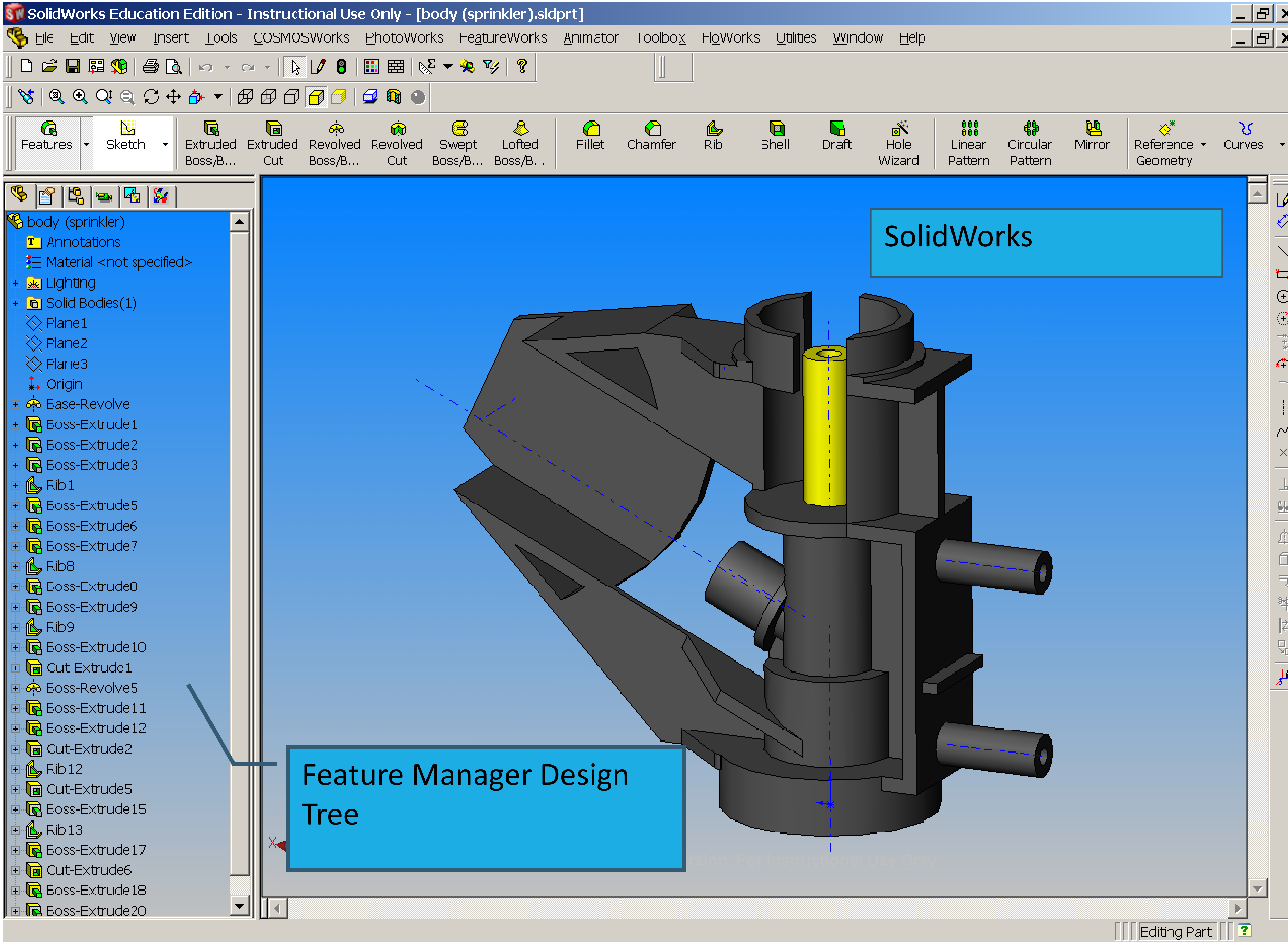
SolidWorks Education Edition - Instructional Use Only - [body (sprinkler).sldprt]
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Features
Sketch
Extruded Boss/B...
Extruded Cut
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Revolved Cut
Swept Boss/B...
Lofted Boss/B...
Fillet
Chamfer
Rib
Shell
Draft
Hole Wizard
Linear Pattern
Circular Pattern
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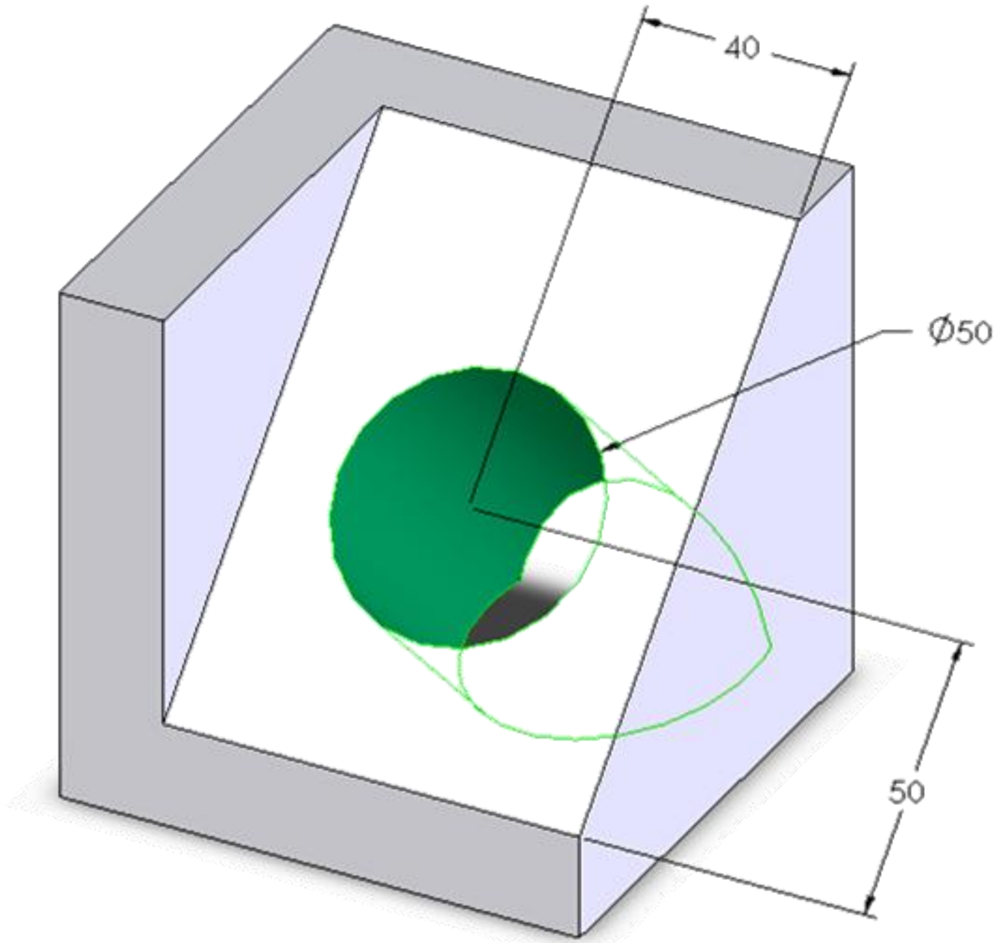
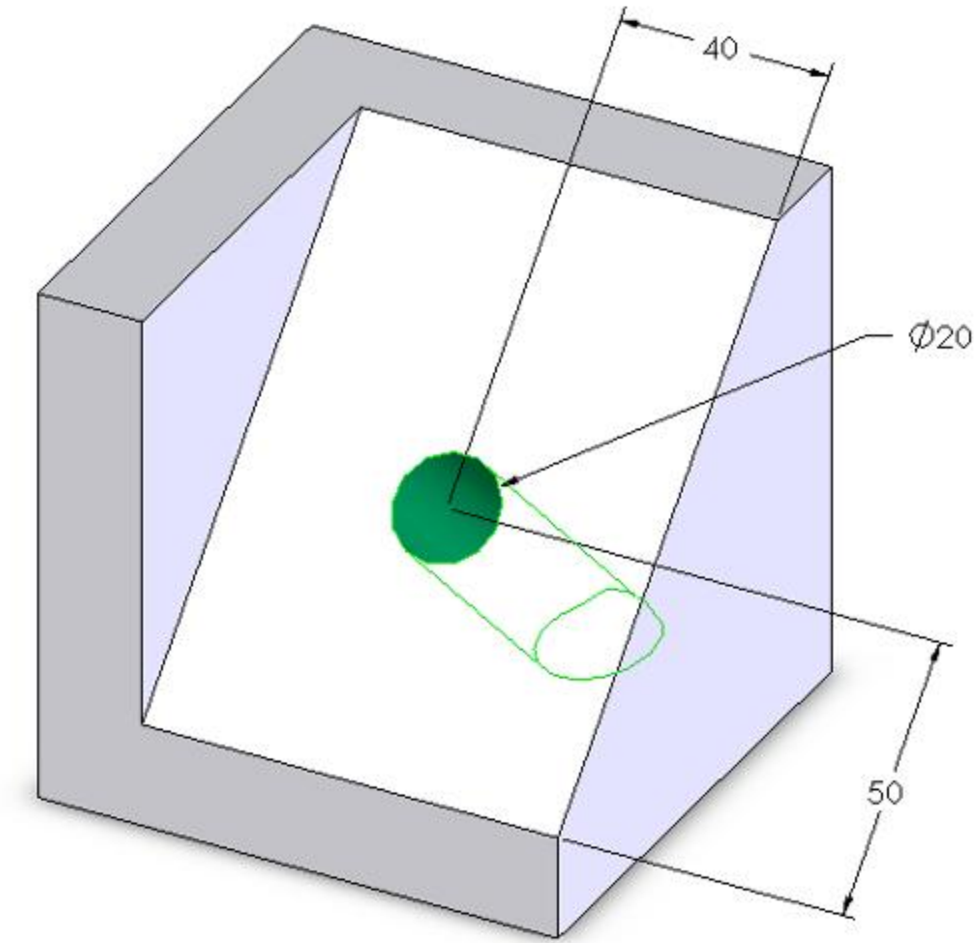
Reference Geometry
Curves
body (sprinkler)
Annotations
Material <not specified>
Lighting
Solid Bodies(1)
Plane1
Plane2
Plane3
Origin
Base-Revolve
Boss-Extrude1
Boss-Extrude2
Boss-Extrude3
Rib1
Boss-Extrude5
Boss-Extrude6
Boss-Extrude7
Rib8
Boss-Extrude8
Boss-Extrude9
Rib9
Boss-Extrude10
Cut-Extrude1
Boss-Revolve5
Boss-Extrude11
Boss-Extrude12
Cut-Extrude2
Rib12
Cut-Extrude5
Boss-Extrude15
Rib13
Boss-Extrude17
Cut-Extrude6
Boss-Extrude18
Boss-Extrude20
Editing Part
SolidWorks
Feature Manager Design Tree

Modifying Parts

- The part is created from the history tree
- Features can be added, deleted and re-ordered
- Feature parameters can be changed



ES1050 part 01.SLDPRT



ES1050 part 01.SLDPRT

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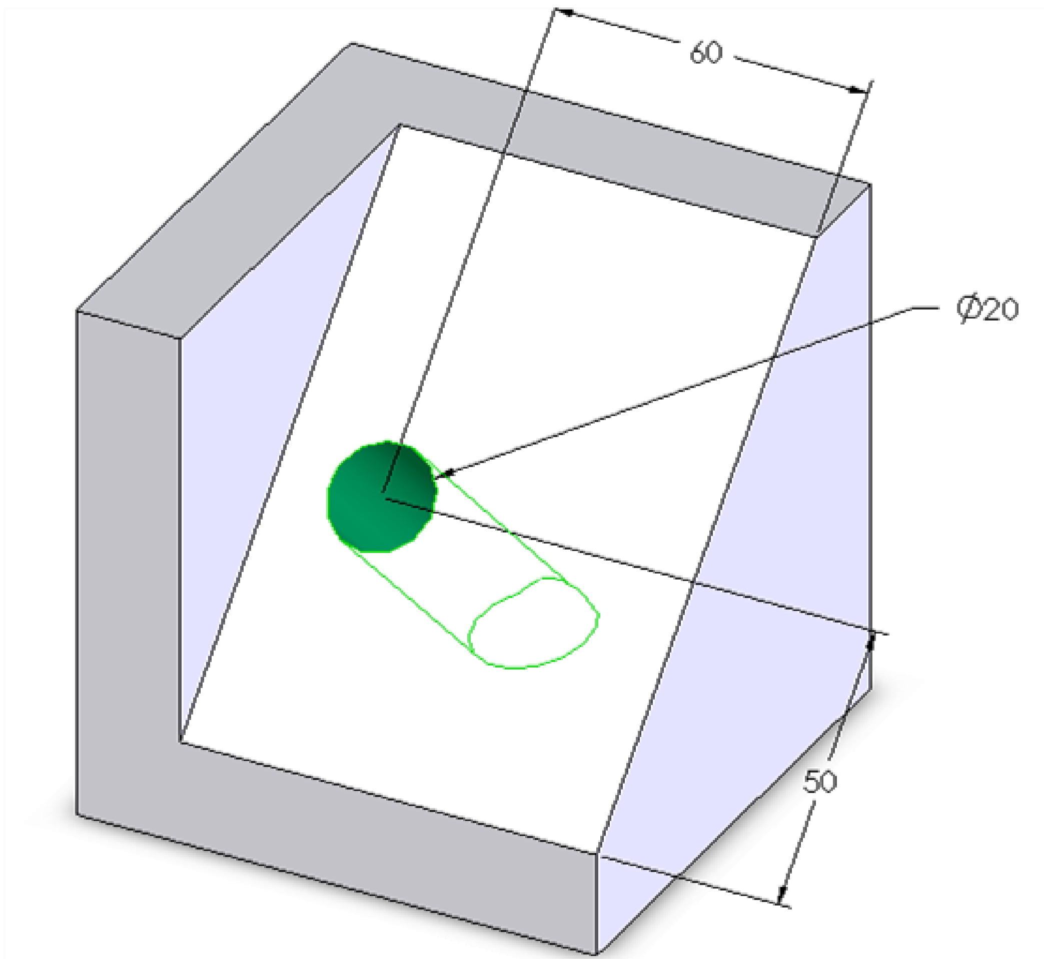


ES1050 part 01.SLDPRT

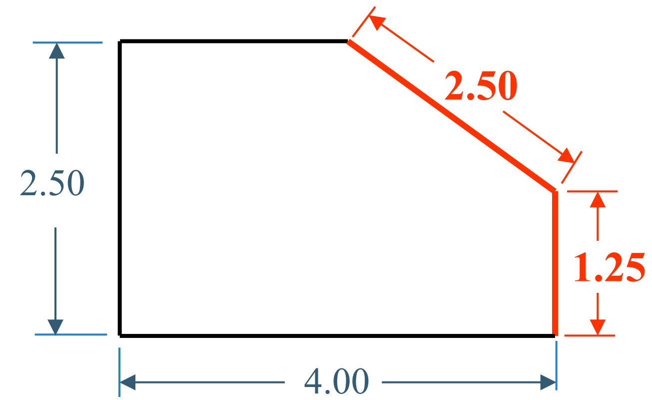
Summary

- Most CAD systems use solid, parametric, feature-based modeling
- Parts are modeled by adding features to a base feature
- Features can be easily added, deleted and modified

Design Intent

- In parametric modeling, dimensions control the model.
- Design intent is how your model will react when dimension values are changed.

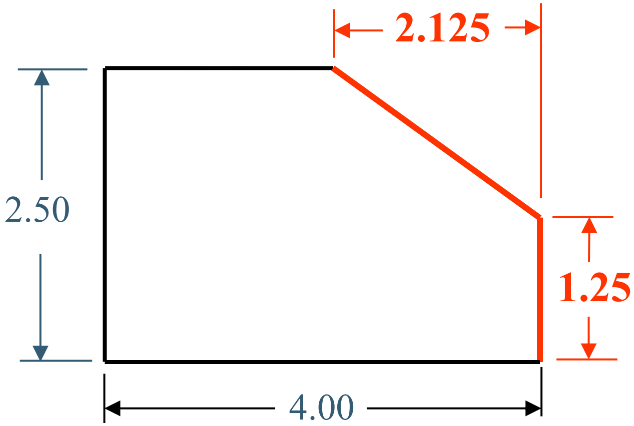
Design Intent

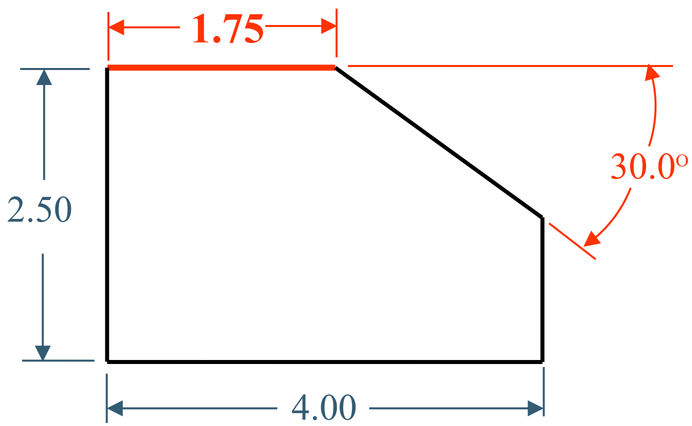


- The drawing shows the intent of the designer that the inclined plane (chamfer) should have a flat area measuring 2.5 inches and that it should start at a point 1.25 inches from the base of the drawing. These parameters are what the designer deemed significant for this model.
- Remember that the placement of dimensions is very important because they are being used to drive the shape of the geometry. If the 2.5 in. vertical dimension increases, the 2.5 in. flat across the chamfer will be maintained, but its angle will change.

Design Intent

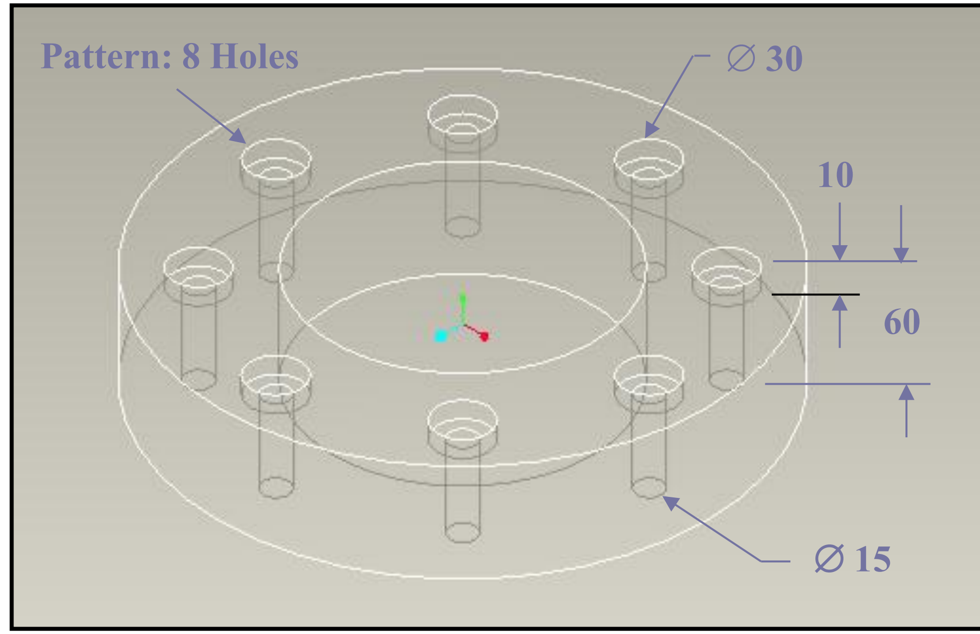
- In this drawing, what is important to the designer is the vertical location and horizontal dimension of the chamfer, rather than the flat of the chamfer.
- In the last drawing, the designer calls for a specific angle for the chamfer. In this case the angle of the chamfer should be dimensioned.





Parametric Modeling

- The true power of parametric modeling shines through when design changes need to be made. The design modification is made by simply changing a dimension.
- Since the counterbore is associated with the top surface of the ring, any changes in the thickness of the ring would automatically be reflected on the counterbore depth.

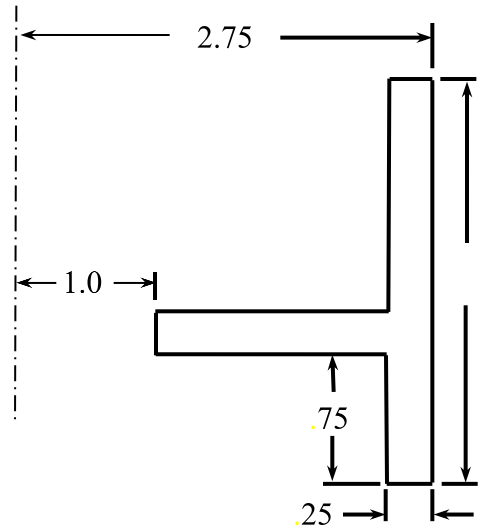


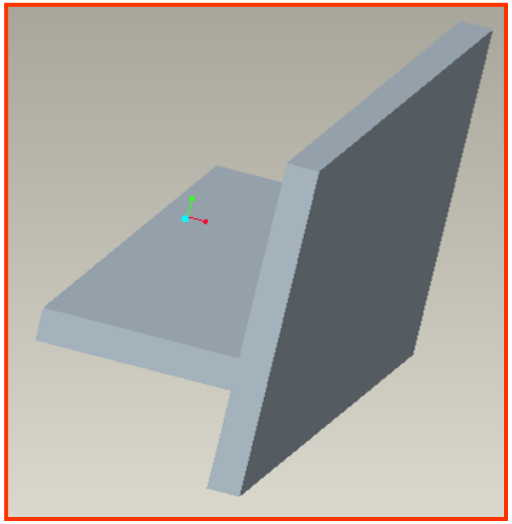
Sketches

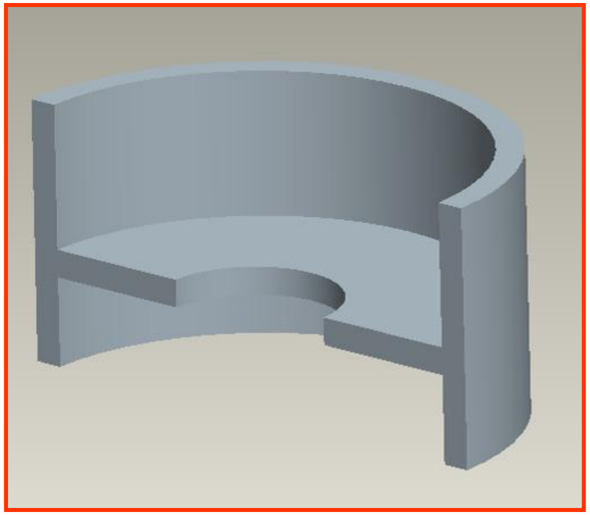
- Take the word sketch literally. A sketch should be just that, a sketch.
- When sketching it is not necessary to create geometry with accuracy. Lines, arcs, and additional geometry need not be created with exact dimensions in mind.
- When the dimensions are added, the sketch will change size and shape. This is the essence of Parametric Modeling.
- In short, the sketch need only be the approximate size and shape of the part being designed. When dimensions are added, they will drive the size and the shape of the geometry.

Features

- Sketched Feature
  - Create a 2D sketch.
  - Create a feature from the sketch by extruding, revolving, sweeping, lofting and blending.

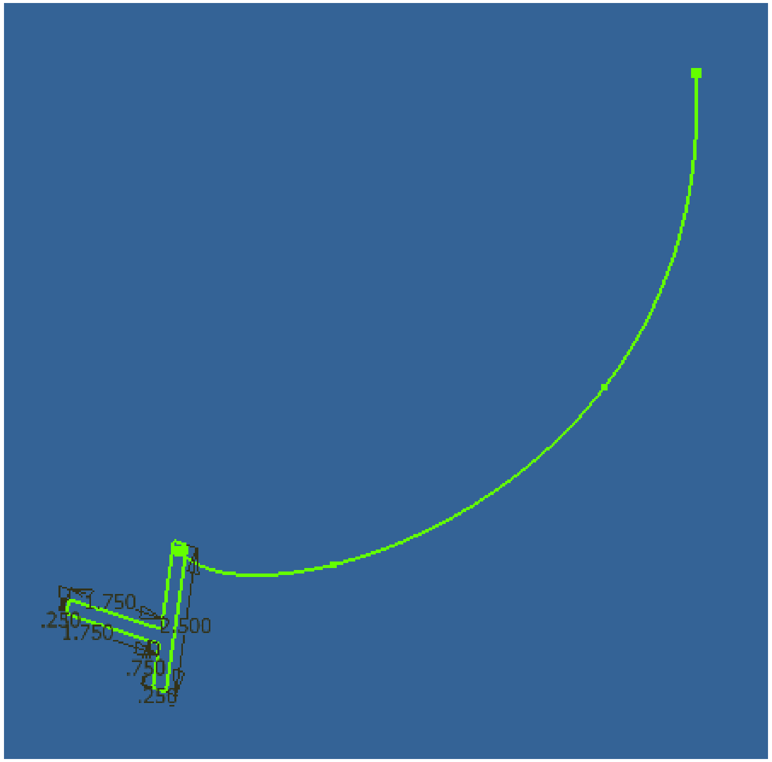


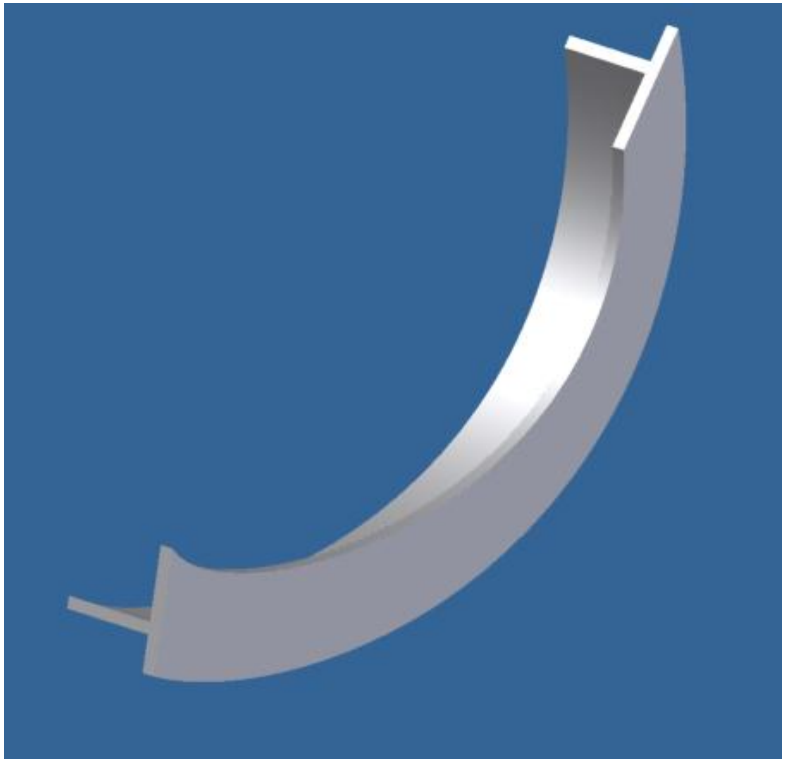




Sweep

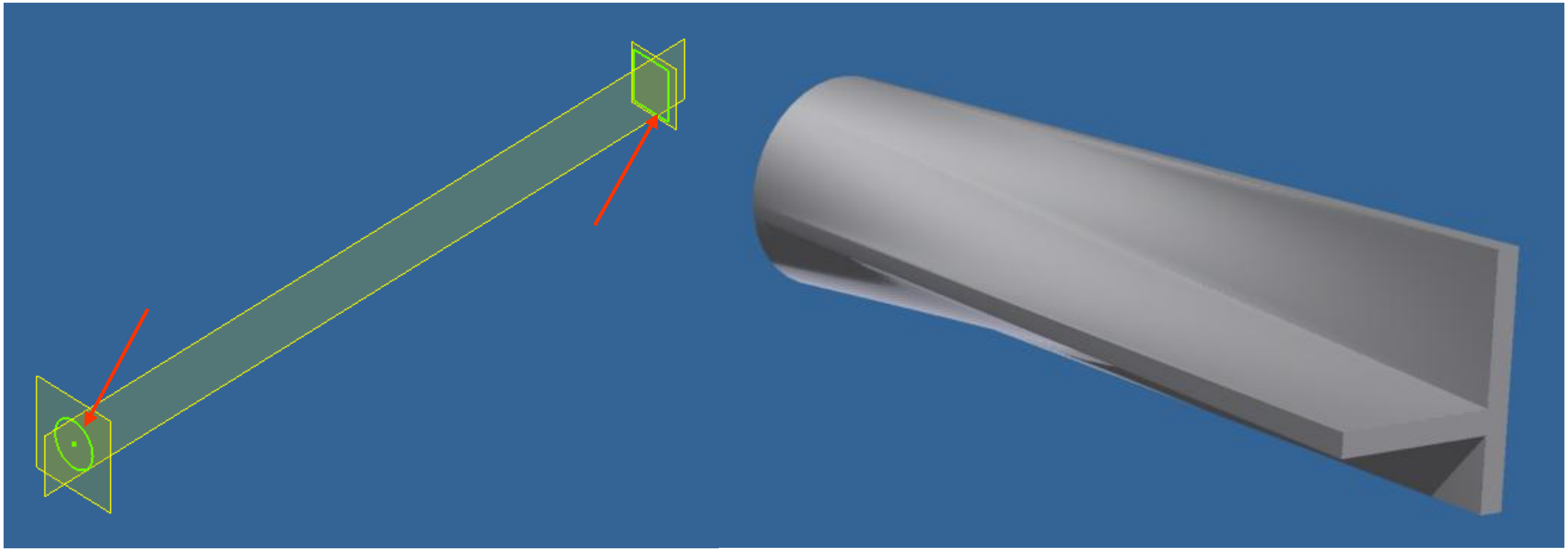
A Sweep feature requires a profile and a path. The profile will follow the path to create the solid.



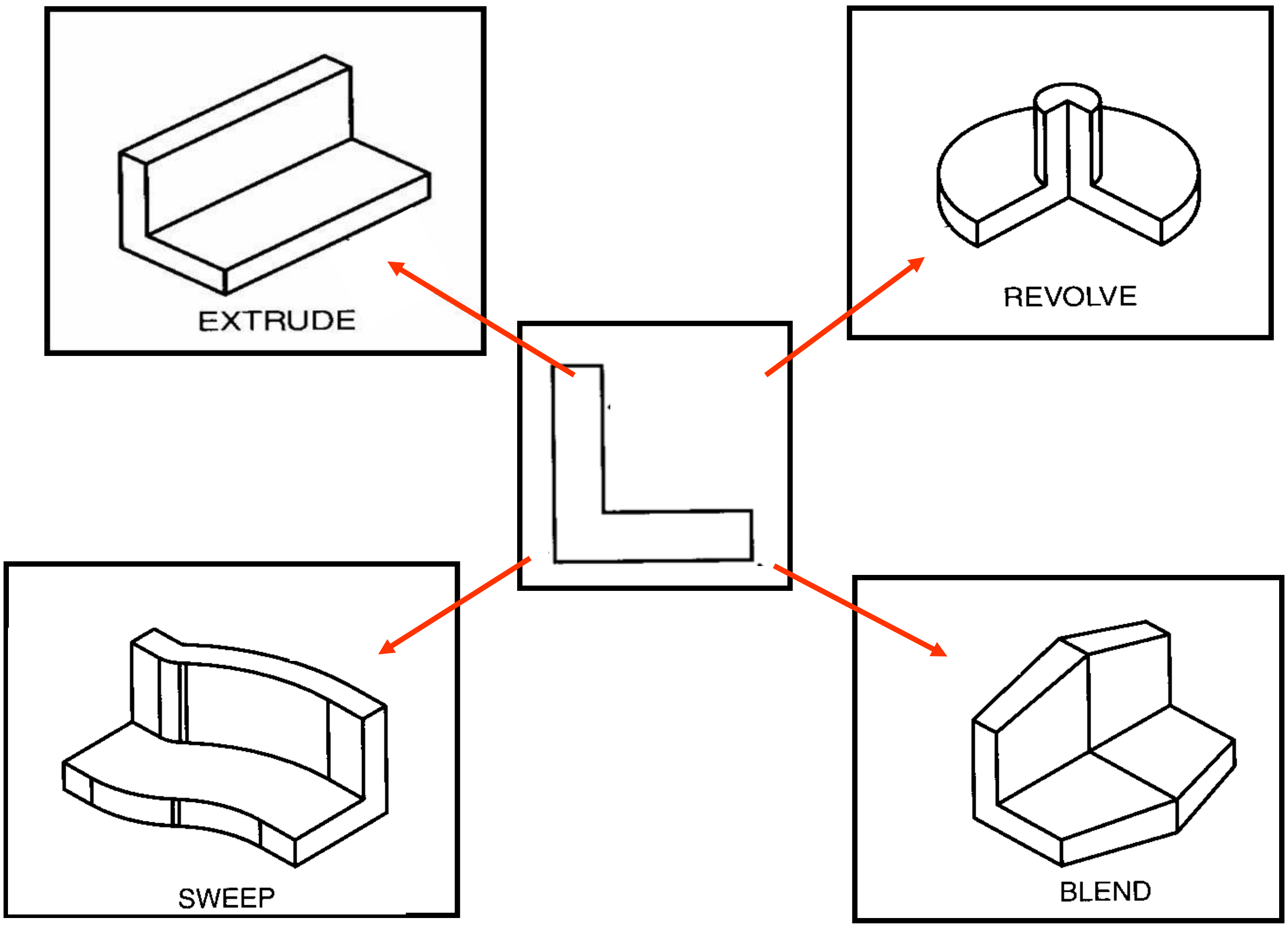


Loft

- Sections (profiles) do not have to be sketched on parallel planes
- All sections must be either closed or open

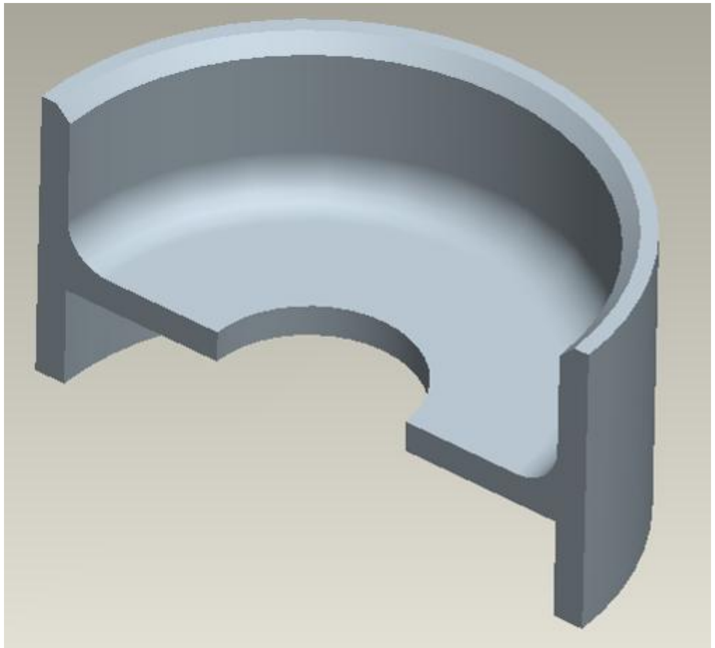


Creating Features from Sketches

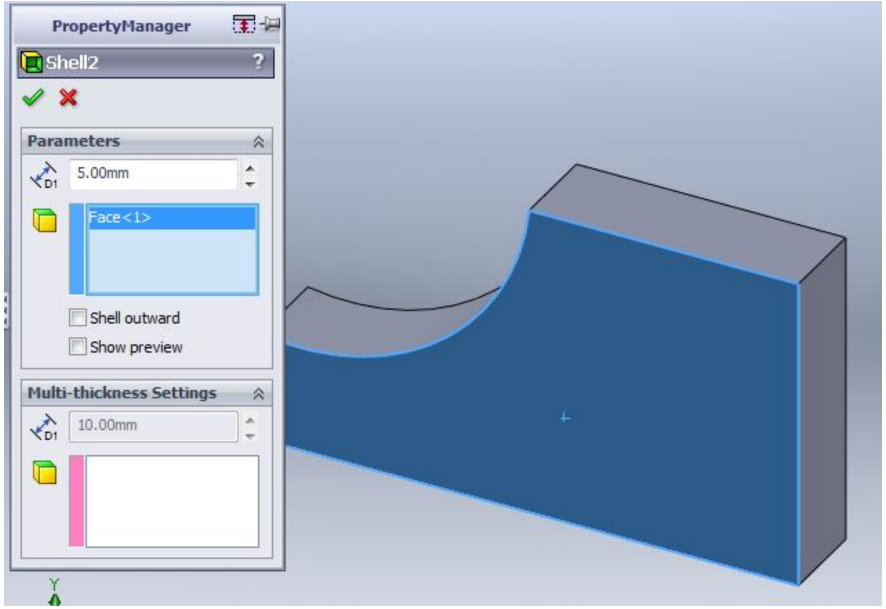


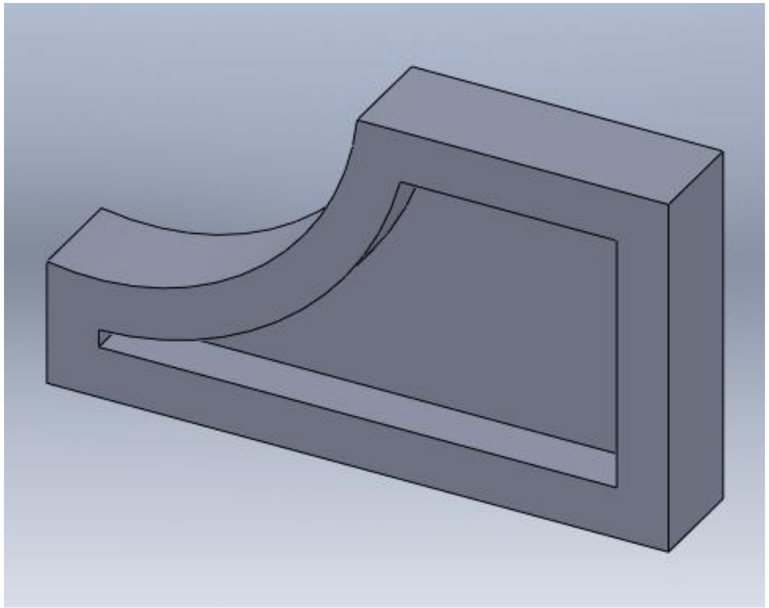
Tweak Features

- Applied feature does not require a sketch.
- It is applied directly to the model.
- Fillets and chamfers are very common applied features.

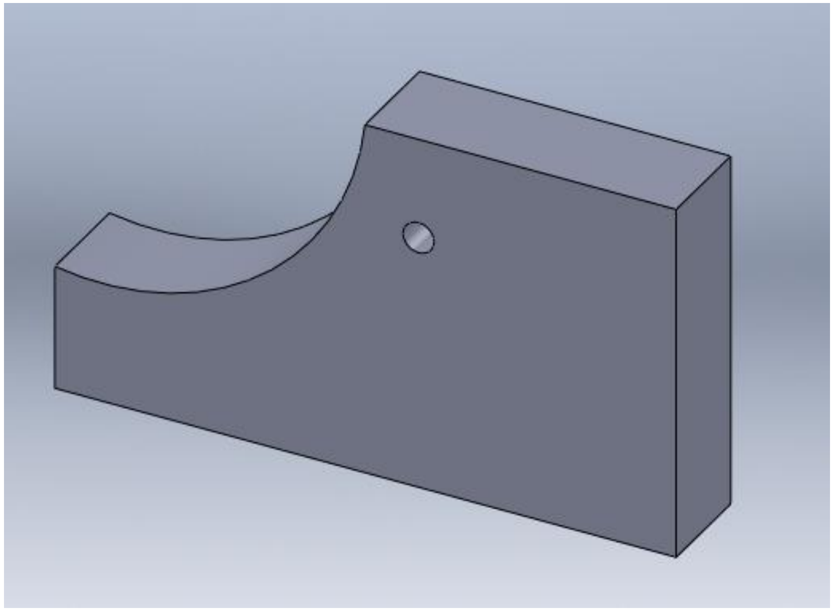


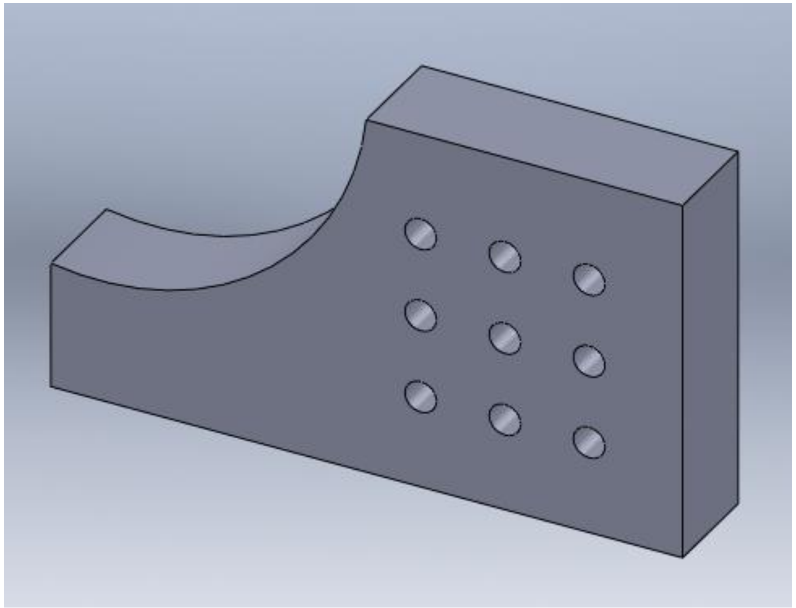
Shell - Hollow

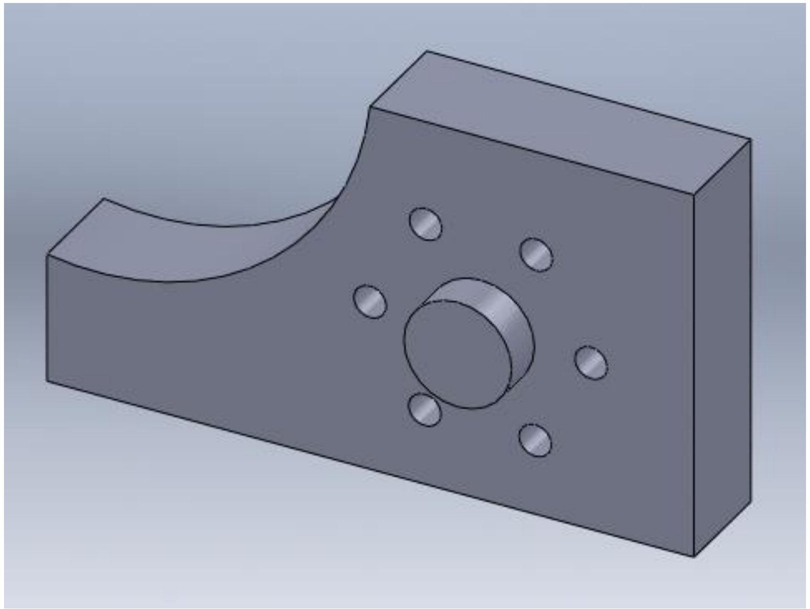




Patterns



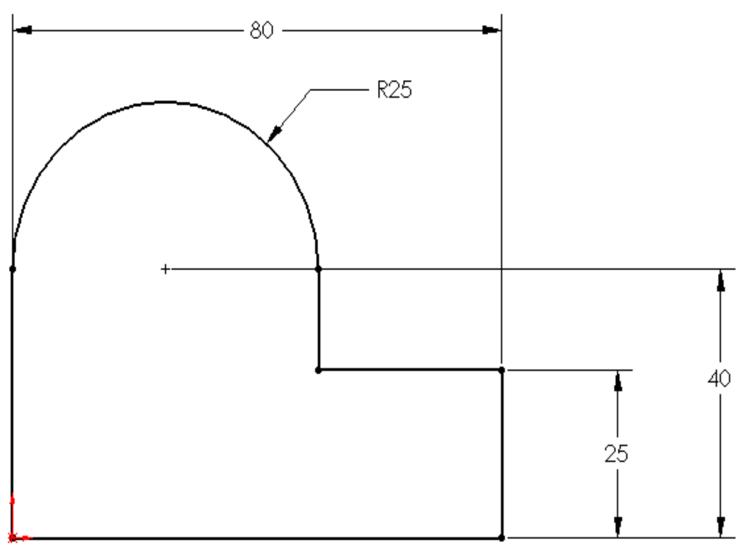


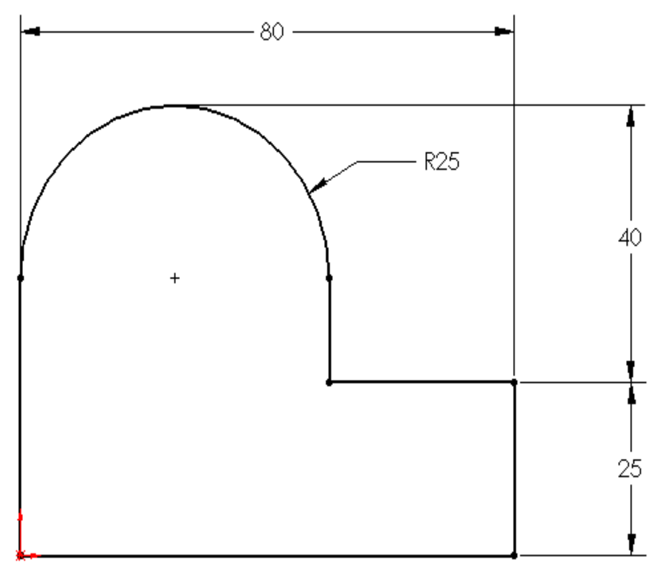


Exercises

Comparing Two Sketches

- Look at the two sketches shown
- Think about how they differ.
- What happens to the overall height when R25 changes to R15?

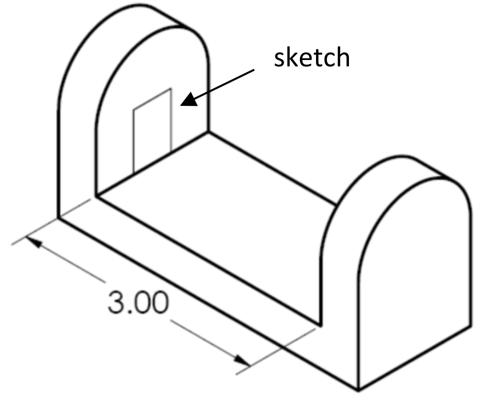




Design Intent

- There are several approaches you can take to create a 3D solid model, but some of them are more useful than others.
- The most effective way to create a parametric model is the one that allows the user to update the model with the least amount of effort and time.
- Design intent is the plan of how the model should behave when modified. It is determined by the function of the part. How will the model react if changes need to be made in the future?

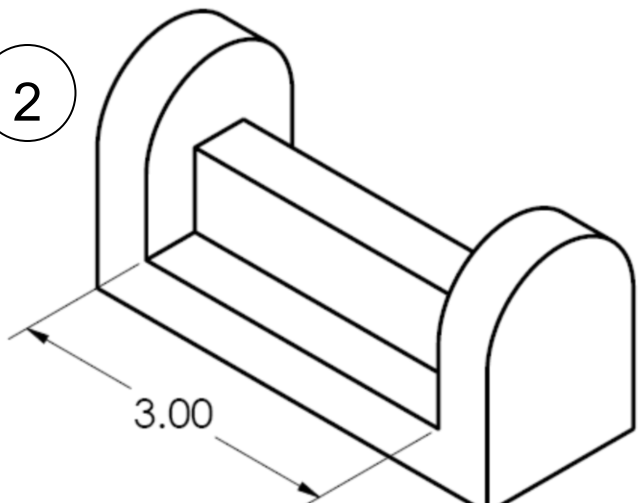
Design Intent
Example



Original Part
and sketch

1

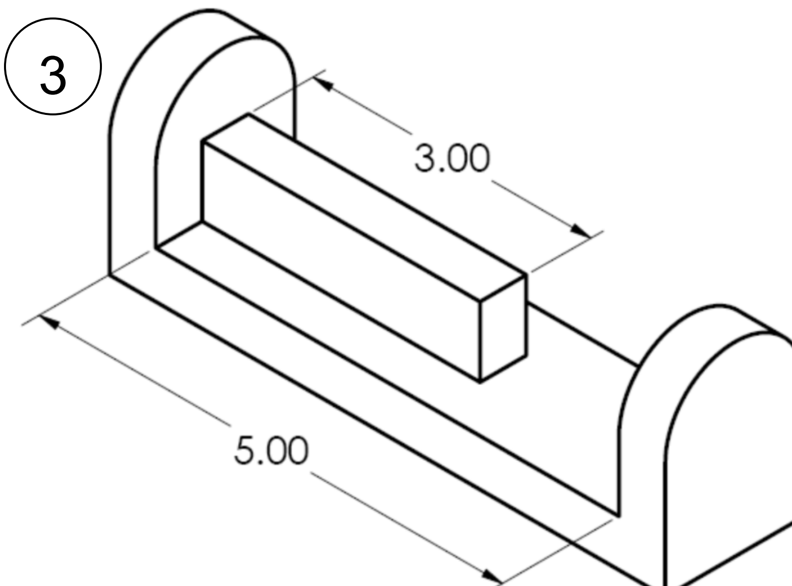
2



Sketch extruded 3”

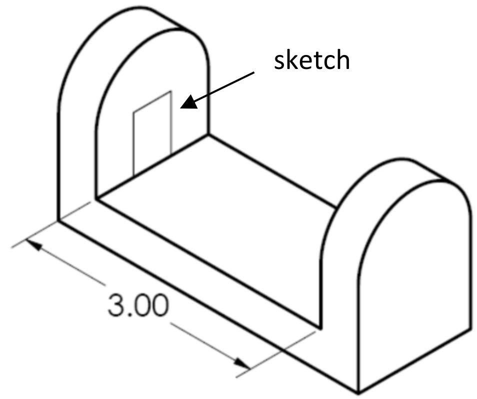
Not efficient!

3



Resultant Part after overall
sketch is changed

Design Intent
Example

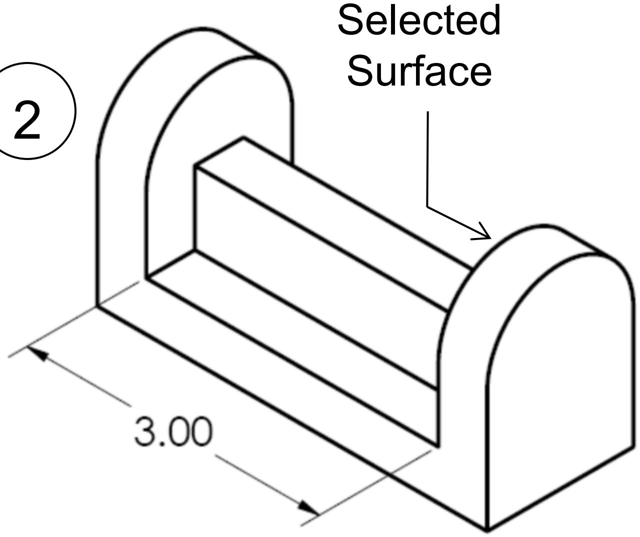


Original Part and sketch

1

2

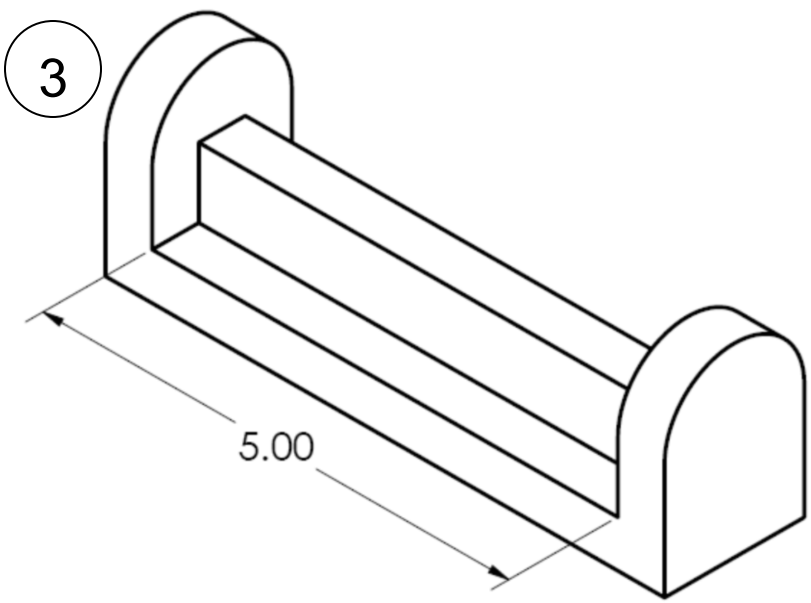
Selected Surface



Sketch extruded “up to existing surface”

Efficient!

3

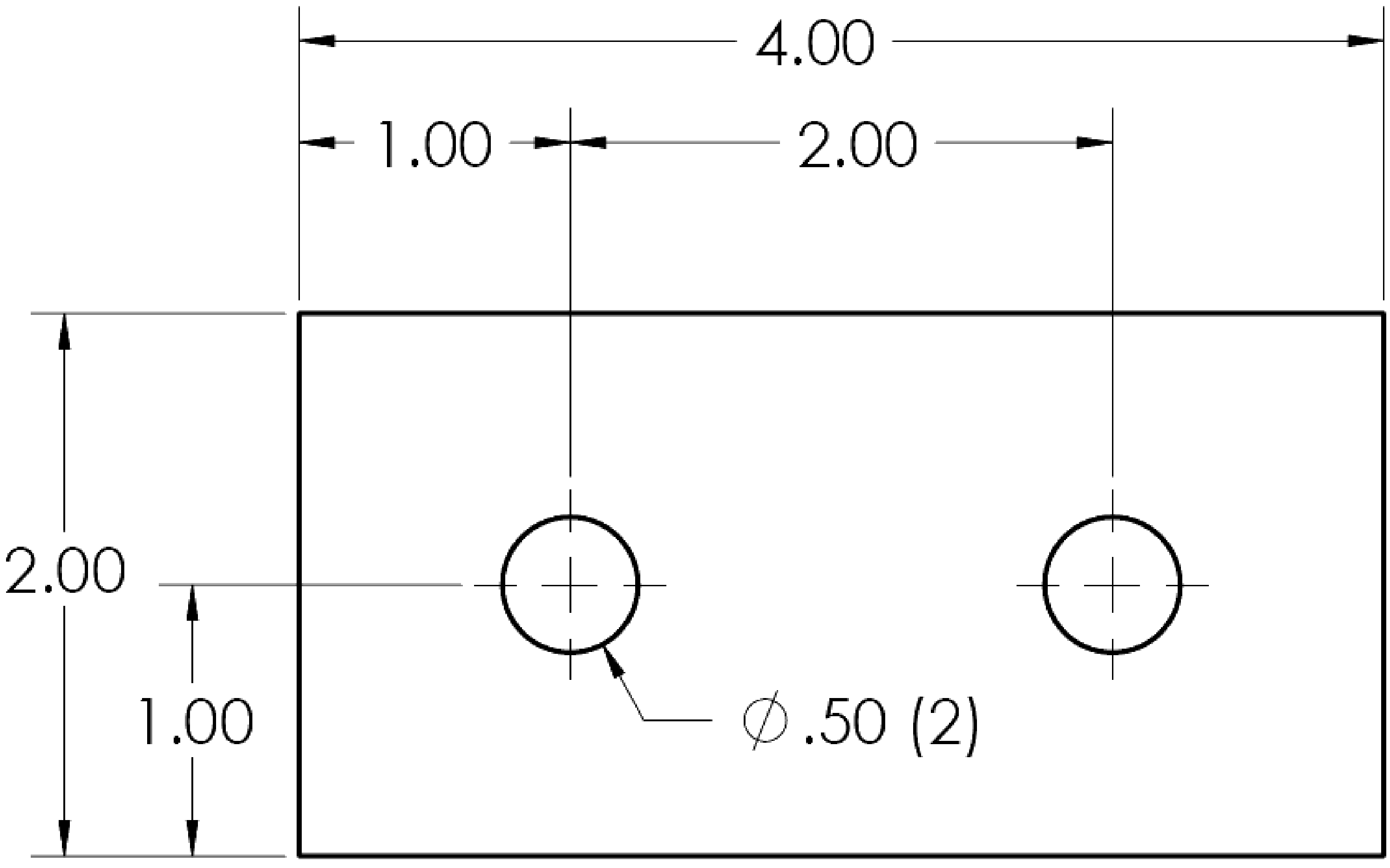


Resultant Part after overall sketch is changed

Design Intent
Implementation

- Capturing design intent is the result of knowledge, experience, skills, and practice. Design Intent is what separates a good model from a bad one.
- Design intent can be implemented in all stages of the parametric modeling process, but the way in which constraints are applied is especially important. Take advantage of geometric constraints and equations.
- Use your knowledge about the design to determine design intent: What features are most likely to change in the future? What condition is the design required to satisfy at all times (equally spaced holes, proportions, etc)?

Design Intent
Example

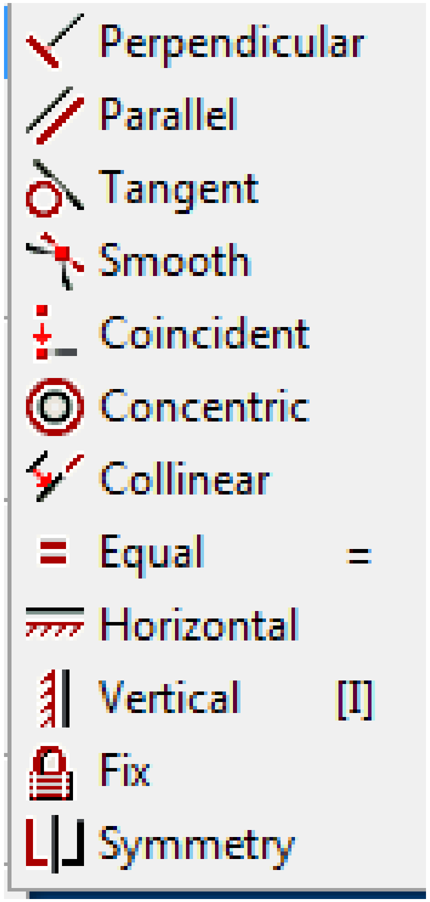


Design Intent
Example

- Let’s change the width of the part to be 5 inches.
- Does the hole pattern look correct?
- Let’s modify the distance between the holes to be 3 inches. Is that better?
- Would it have mattered if we had applied two 1.00 dimensions, rather than a 1.0 and a 2.0?

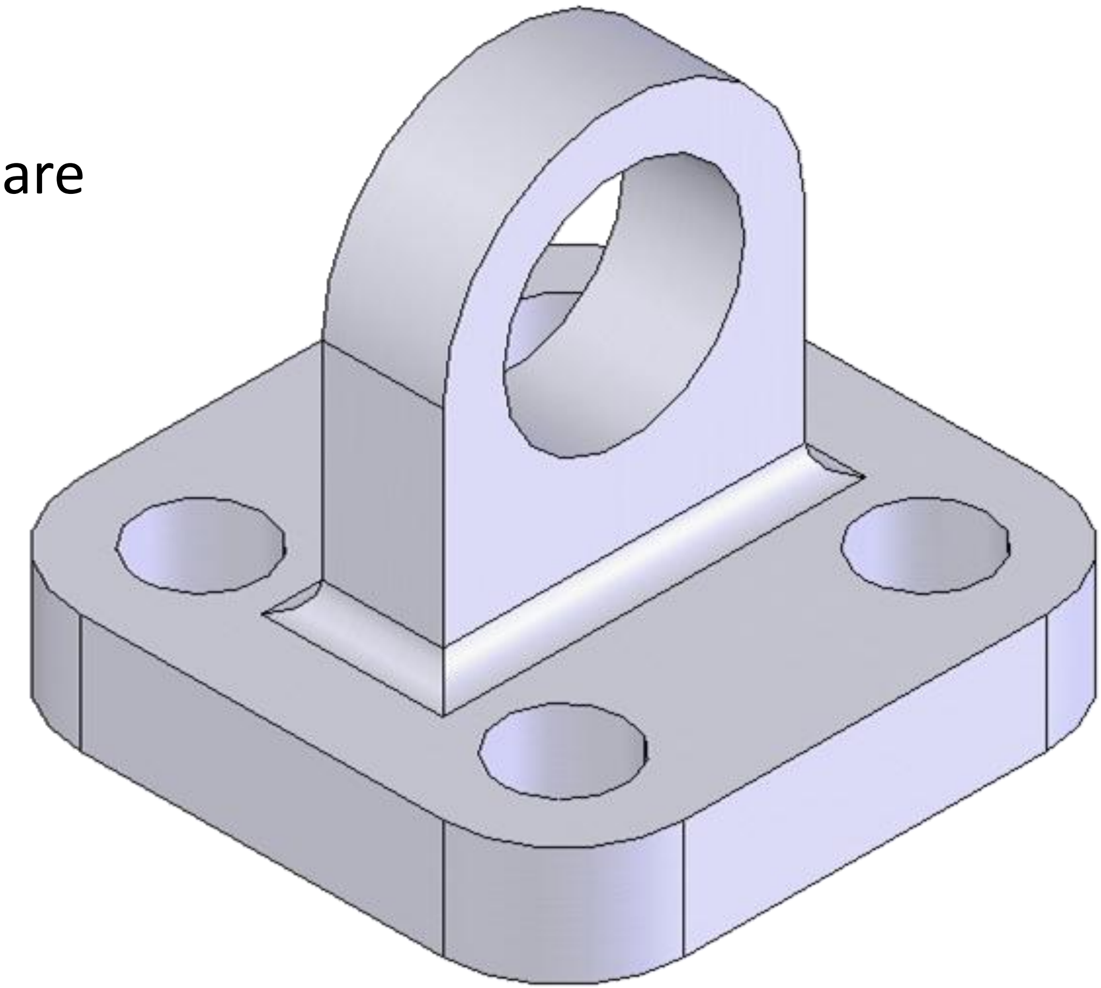
Geometric Constraints

- As you already know, some geometric relationships can be created automatically whenever you sketch an object.
- Other relationships can be added manually.
- The following figure shows common geometric relations.



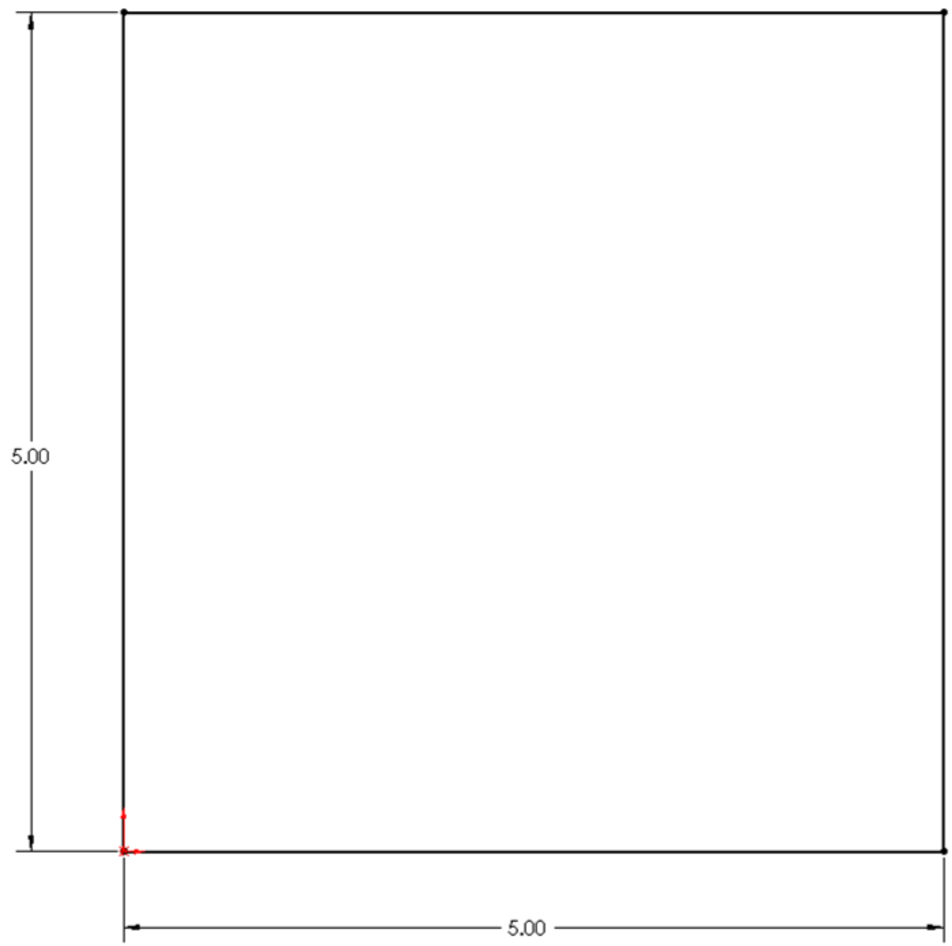
In-class Exercise 8.2.1

- Let's practice!
- This is the object we are going to model.



Exercise 1
First Sketch

- Create a sketch of a square, on the top plane (XZ plane)
  - dimensions: 5 x 5 in.



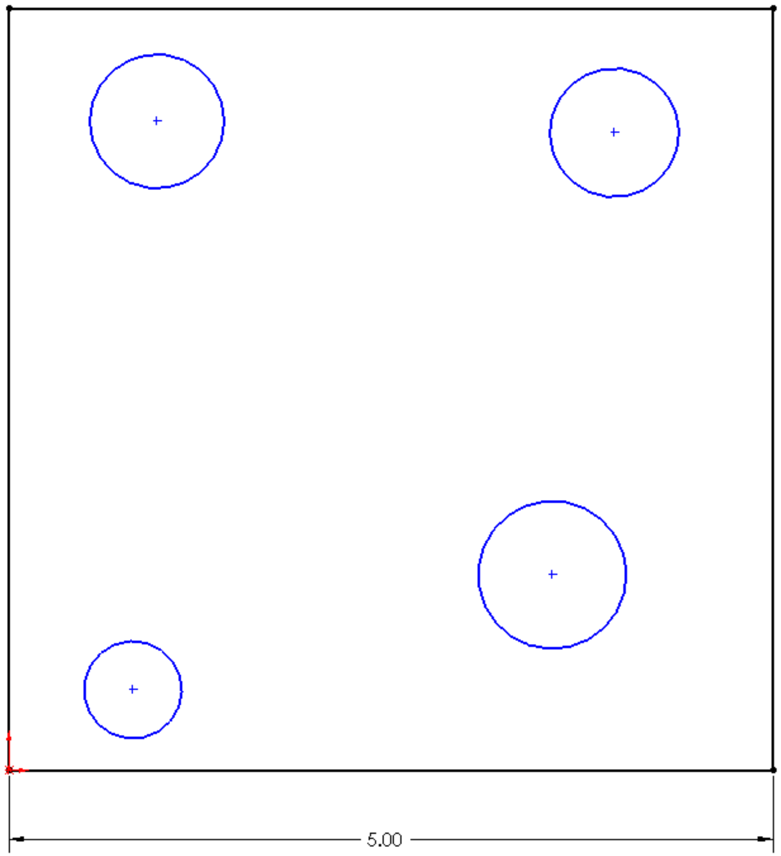
Exercise 1

Keeping the base square

- Rather than applying both 5 inch dimensions, there is another way to fully define the base which requires only 1 dimension (and a geometric constraint).

Exercise 1
Adding the Circles

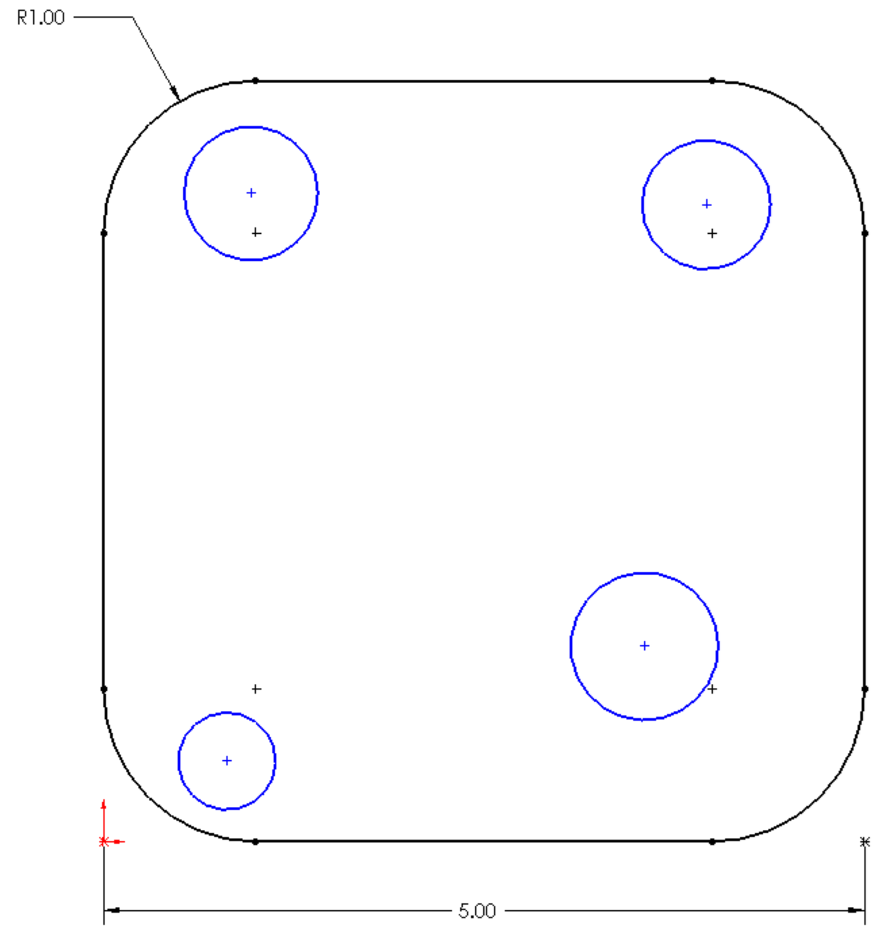
- Add 4 circles to the sketch.
- For now, do not worry about location or size.
- Just place them generally as shown to the right.



Exercise 1

Adding Sketch Fillets

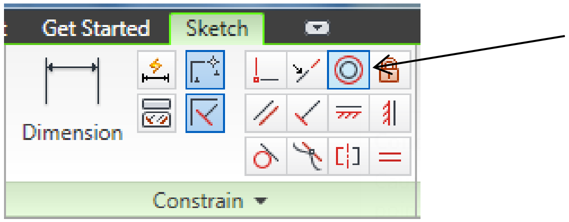
- Add a Sketch Fillet to each of the four corner (Radius=1”)

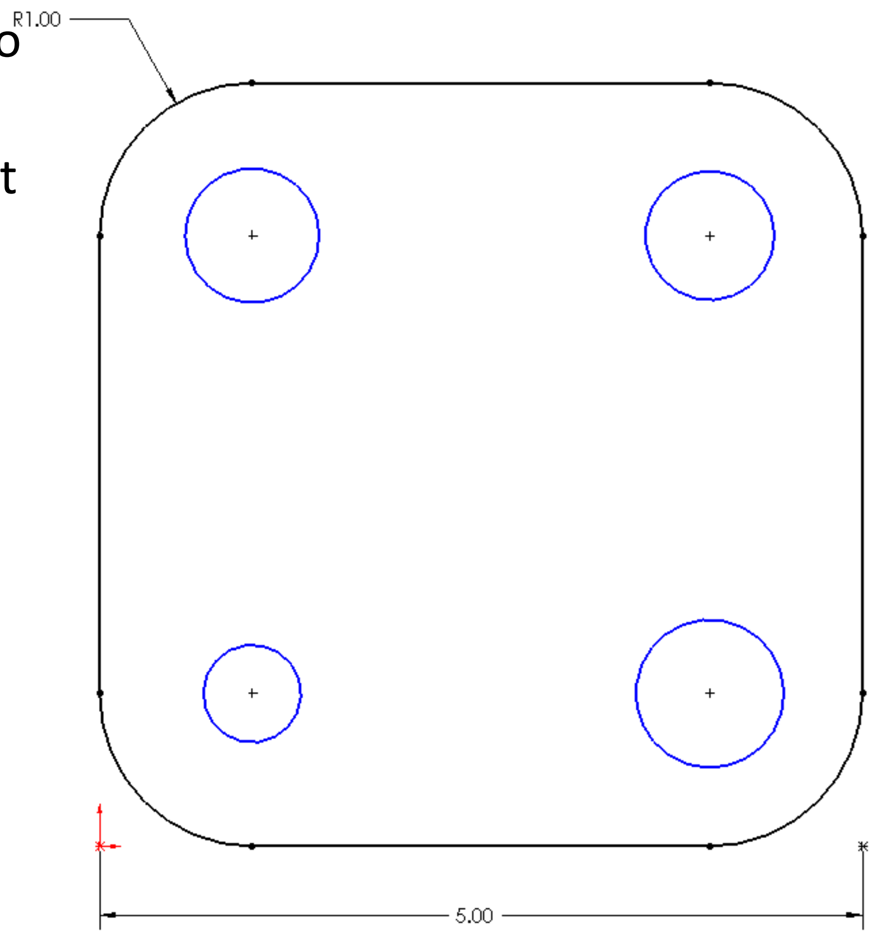


Exercise 1

Adding Geometric Constraints

- Make each of these circles concentric to the corners (rounds).
- Select a rounded corner and then select (holding ctrl) the corresponding circle.
- Now select the concentric relations button in the constrain panel.
- Repeat for the other circles.

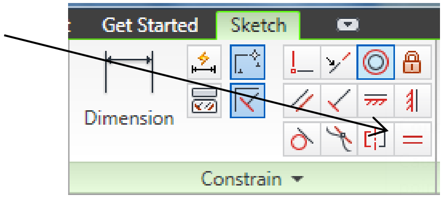


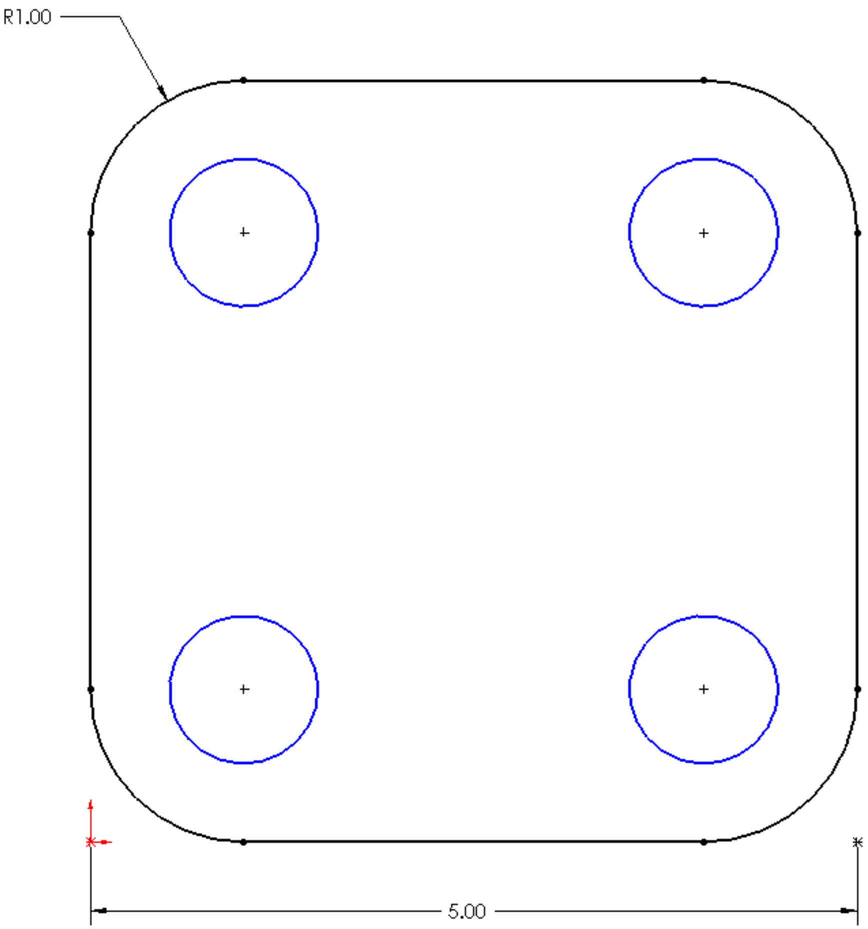


Exercise 1

Making the Circles Equal

- Ctrl -Select all 4 of the circles.
- Select the equal relations button.
- With all of the relationships we have added to our sketch, you will notice that the circles are still (under-constrained).



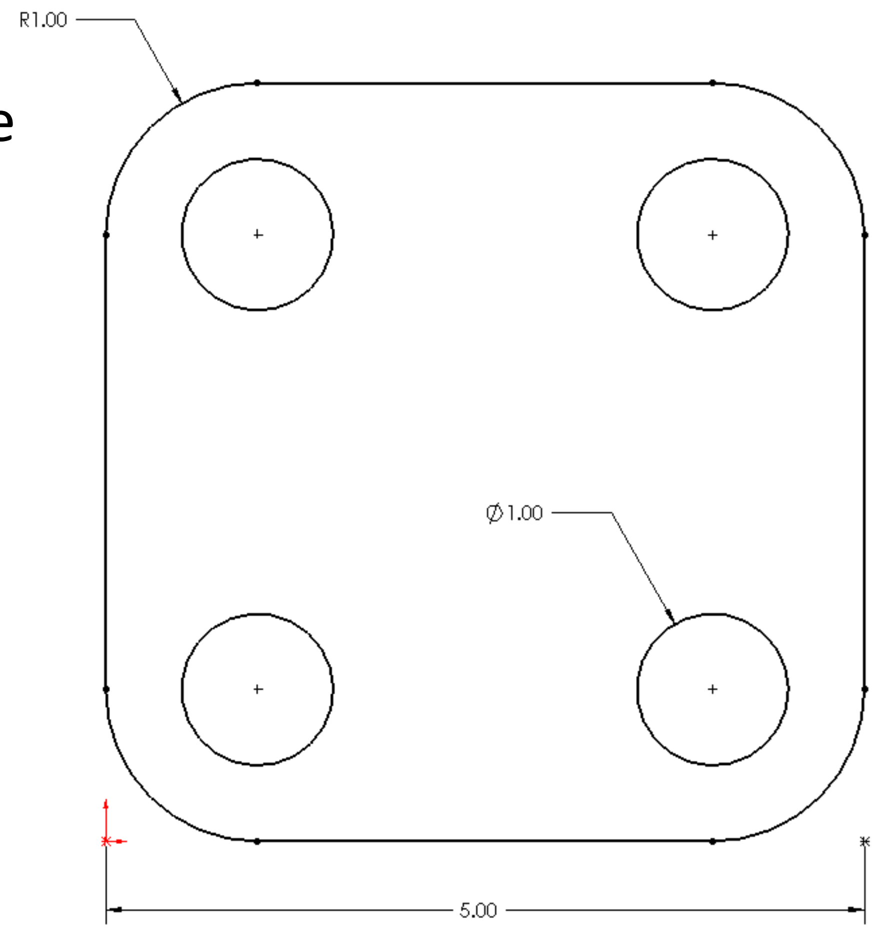


Exercise 1
Fully defining the Sketch

- You will notice at the bottom of the screen it says 1 dimension needed

- Pick a circle and dimension it to a diameter of 1”.

- You will see all of the circles adjust and turn color. Your sketch is now fully constrained.



Exercise 1

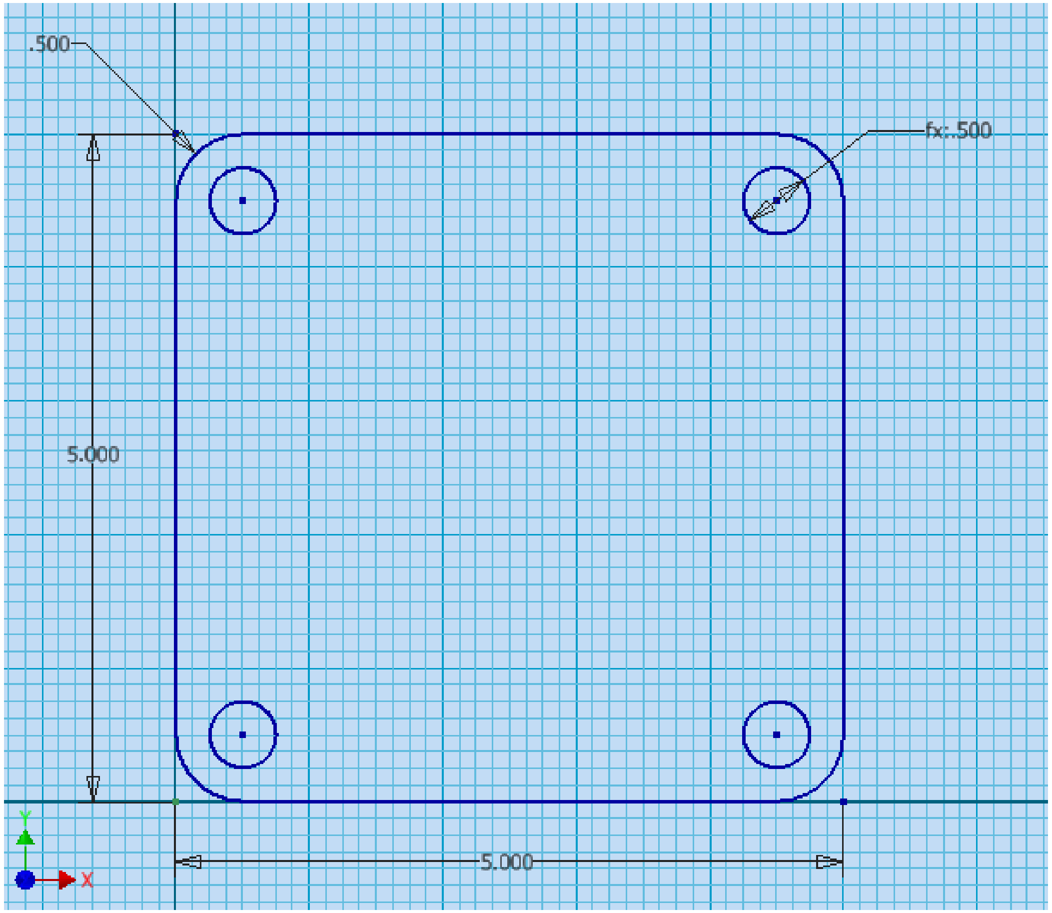
Making it better. Relating the circles and the arcs

- You can use equations to make two (or more) dimensions equal to one another.
- While in dimensioning mode
- Select the 1.000” Radius dimension and type in Radius=1.000 (in the edit box)
- Now select the 1.000” Diameter dimension and type in

  Diameter=Radius (in the edit box)
- The Diameter dimension can now be changed by changing the Radius dimension value
- Try it by changing the radius dimension to .5 and see what happens.

Exercise 1

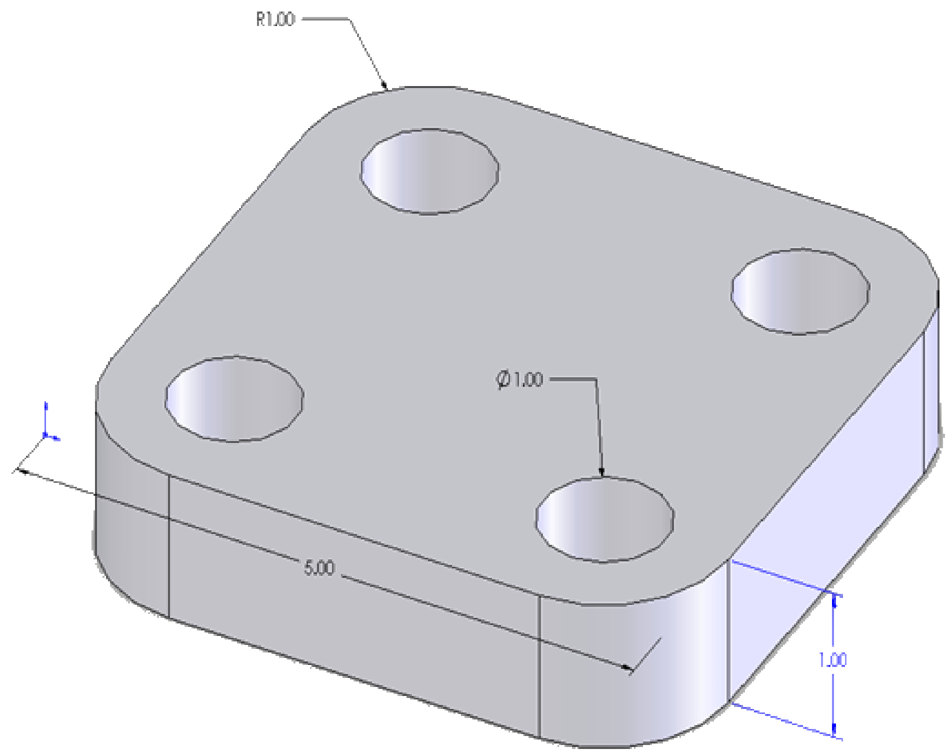
Making it better. Relating the circles and the arcs



Exercise 1

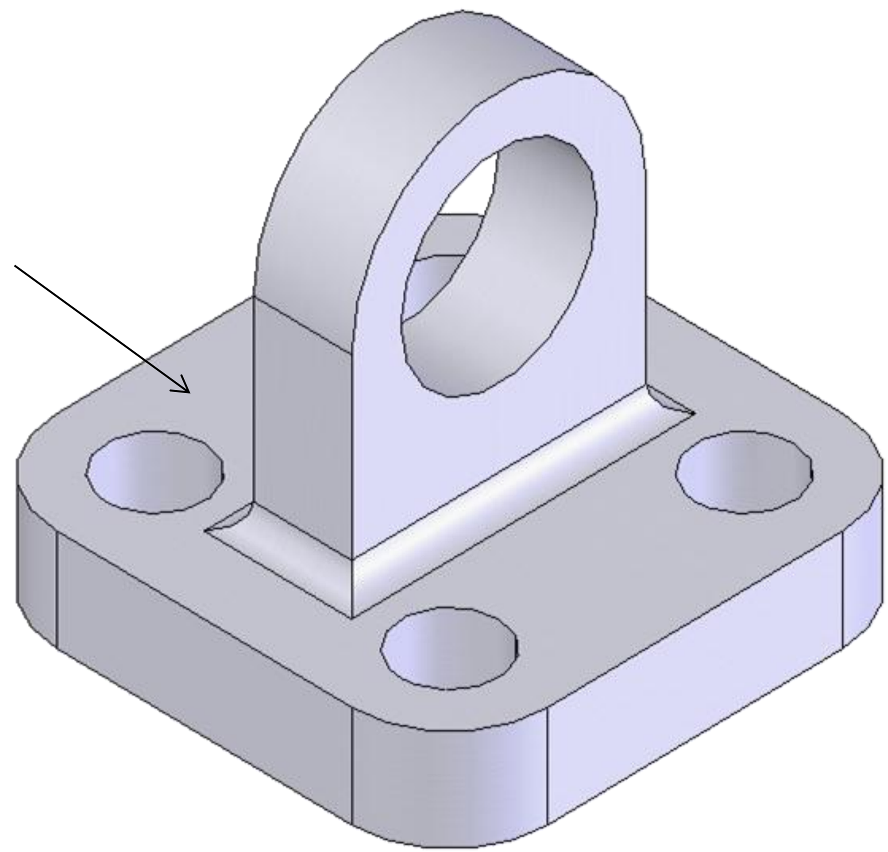
Extruding the base

- Change the radius dimension back to 1.000”
- The thickness of the base is 1”.(extrude)
- All Fillets are 1” radius.
- Holes have a diameter equal to the arc radius and are concentric with them.



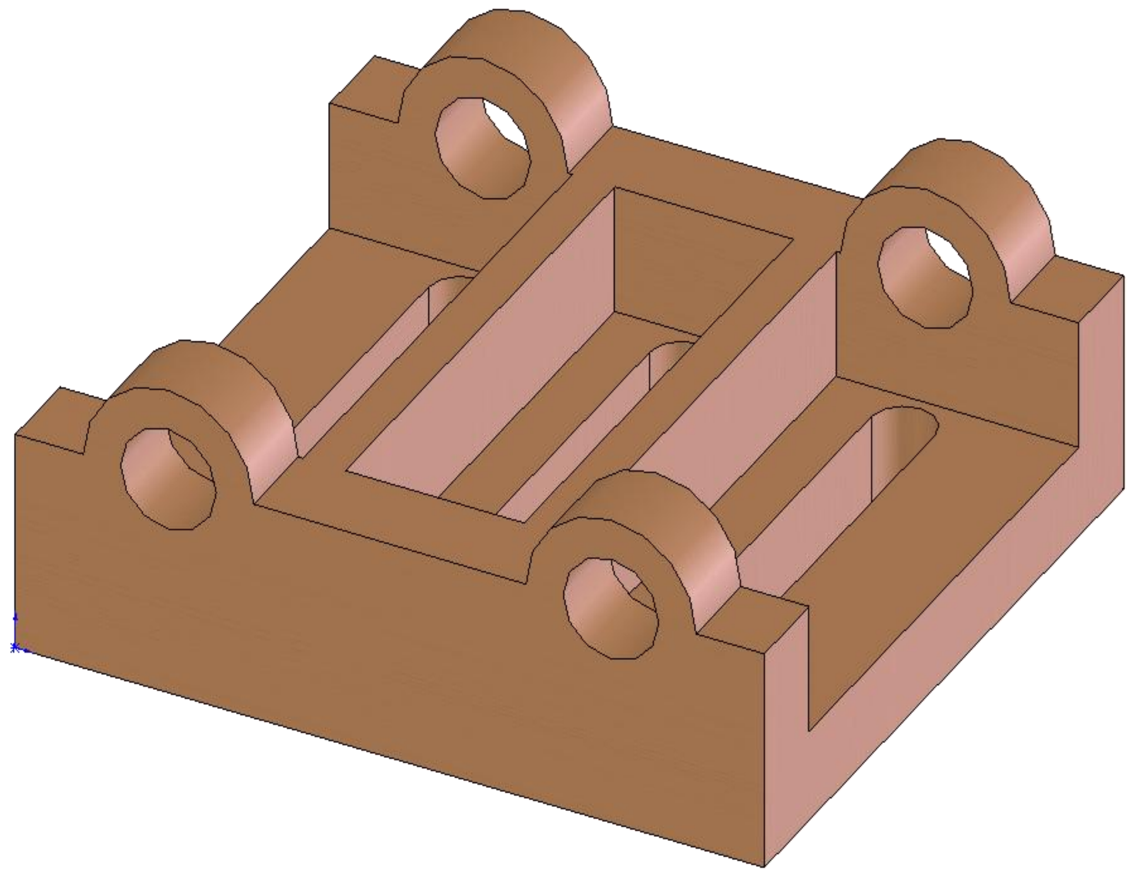
Exercise 1
Adding the Lug

- The Lug has a radius of 1.5 inches and there is 2 inches from the top base surface to the center of the arc.
- Lug thickness is 1.00”
- The hole diameter is 2.00”.
- The fillet is radius .25”
- Design intent – Lug should stay centered when the plate is made larger

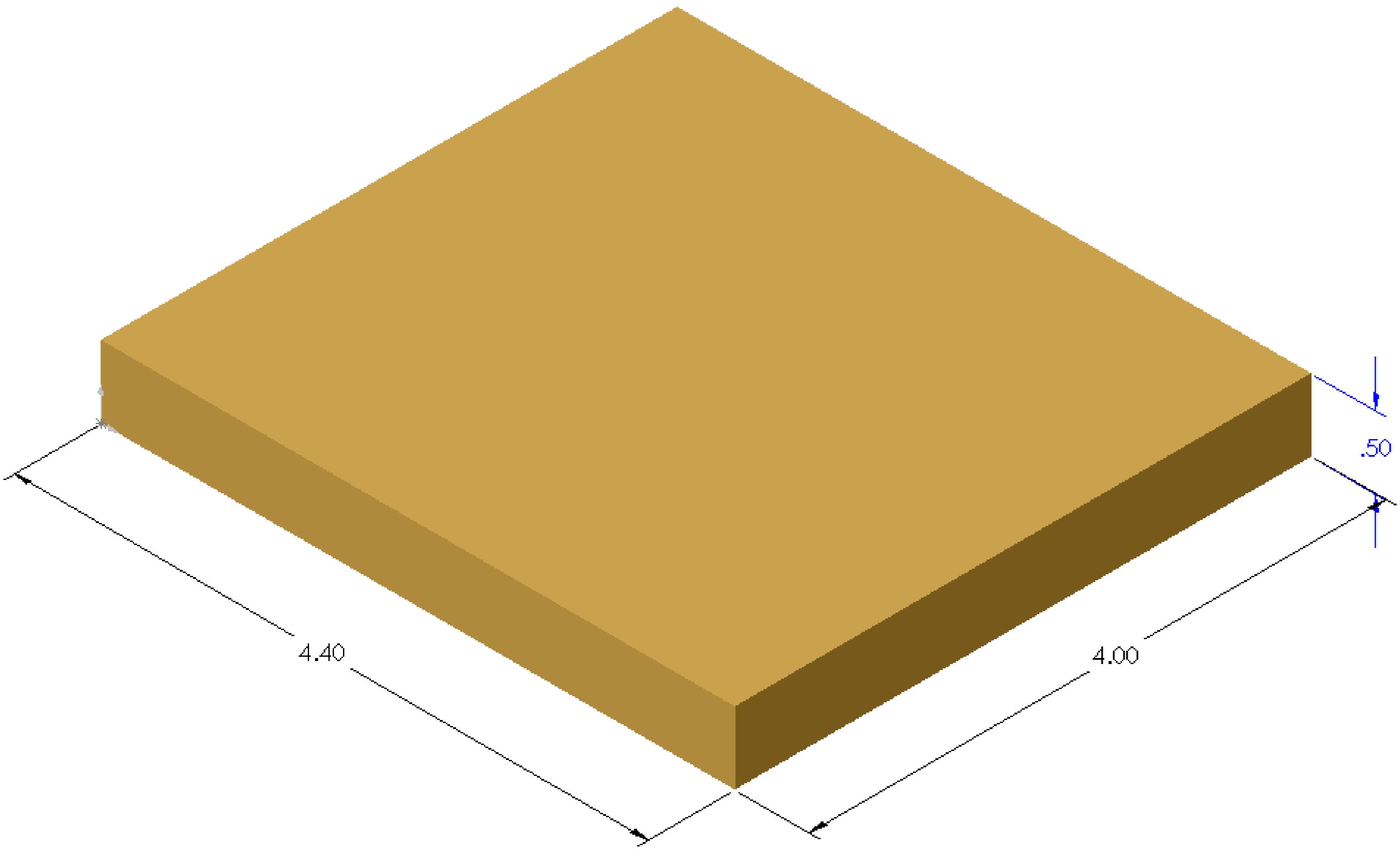


In-class Exercise 8.2.2

- Let’s explore some of the 3D tools in Inventor
- This is the object we are going to model.



Exercise 2
Create base (Dimensions of the base)



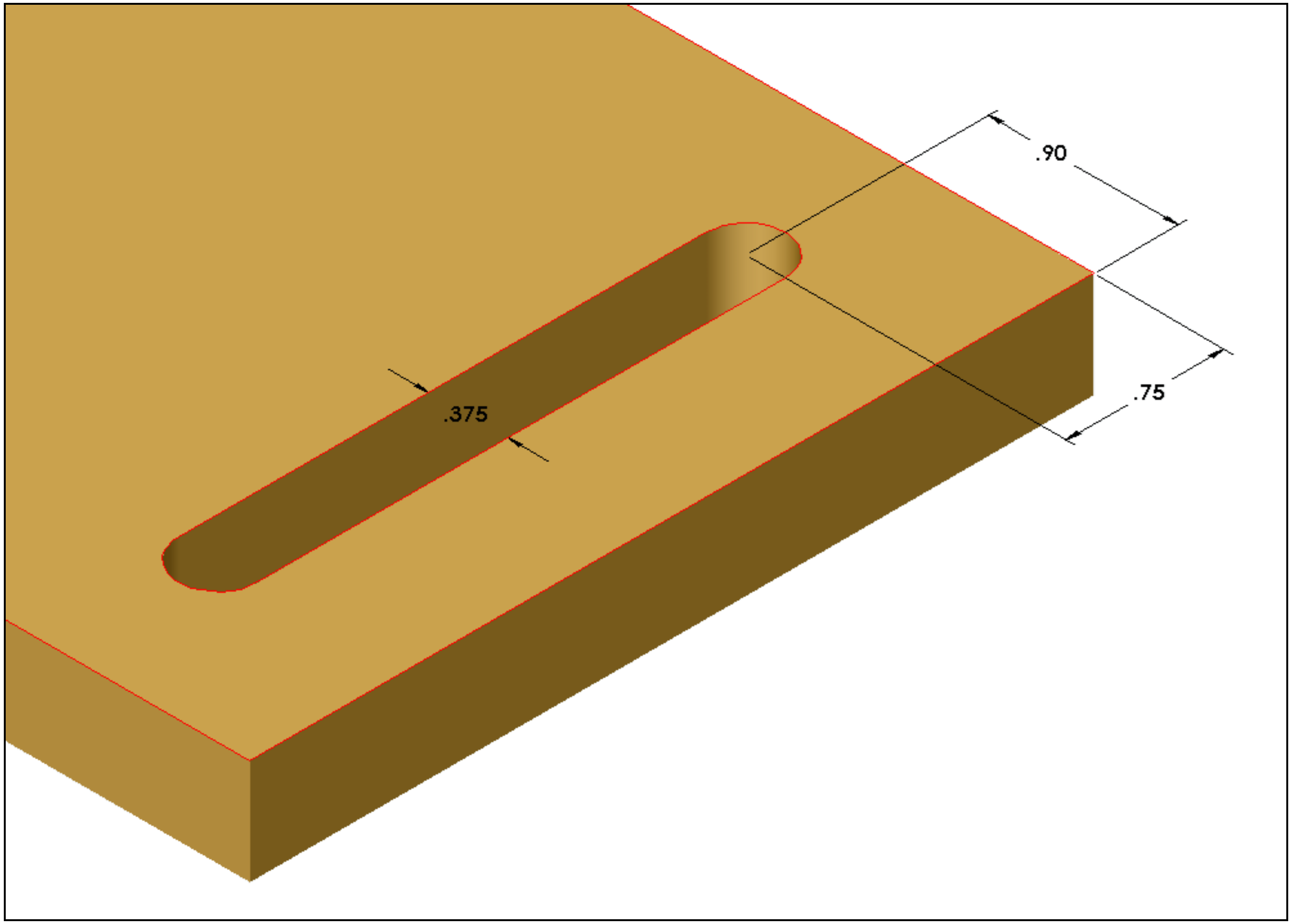
Exercise 2

Creation of the slots

- Create a new sketch on the top surface of the base part.
- Sketch the feature shown in the next slide and Cut it out of the current base part.
- The slot should run the shorter length of the part.

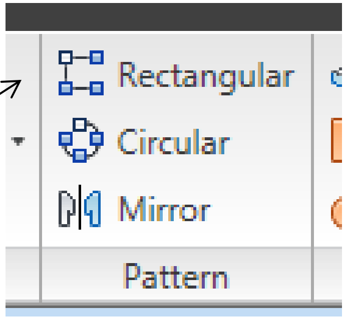
Exercise 2

Creation of the slots (dimensions are typical)



Exercise 2

Creation of the slots.
Using Rectangular Pattern

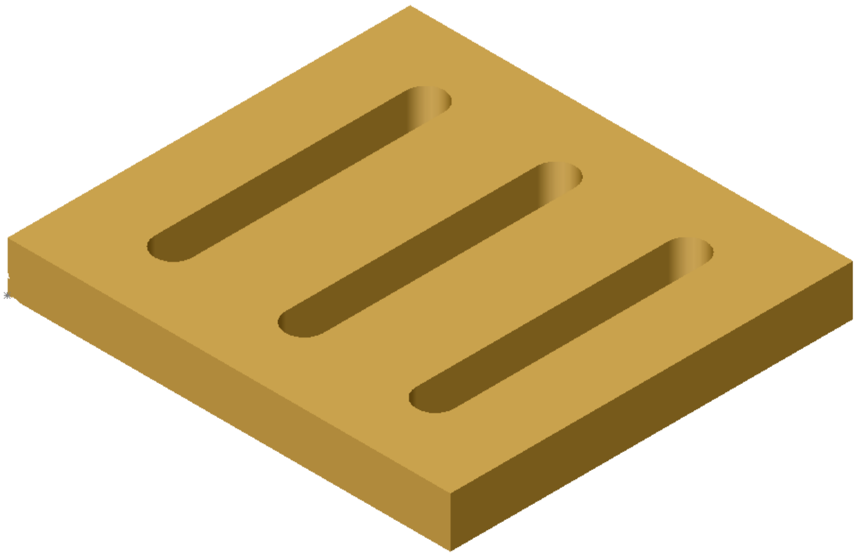


- Once the first slot has been cut, you can use the Rectangular Pattern Command to array a feature or a series of features linearly.
- Step 1:
  - Select the 1st direction you want your pattern to go by picking a linear line or edge. (may have to flip the direction)
  - Select the Distance (Spacing) between each of your instances
  - Select the number of instances (copies).
- Step 2:
  - If the pattern is going to be a 4x4 or 2x3 or multi-directional pattern select the second direction you want to go.
  - Follow the same items as above.
- Step 3:
  - Select the features you want to pattern.
  - These can be selected by clicking on them or selecting them from your part features list.
  - If a feature you created has a fillet or something else placed on it, you need to include those features as well.

Exercise 2

Creation of the slots. Using Rectangular Pattern

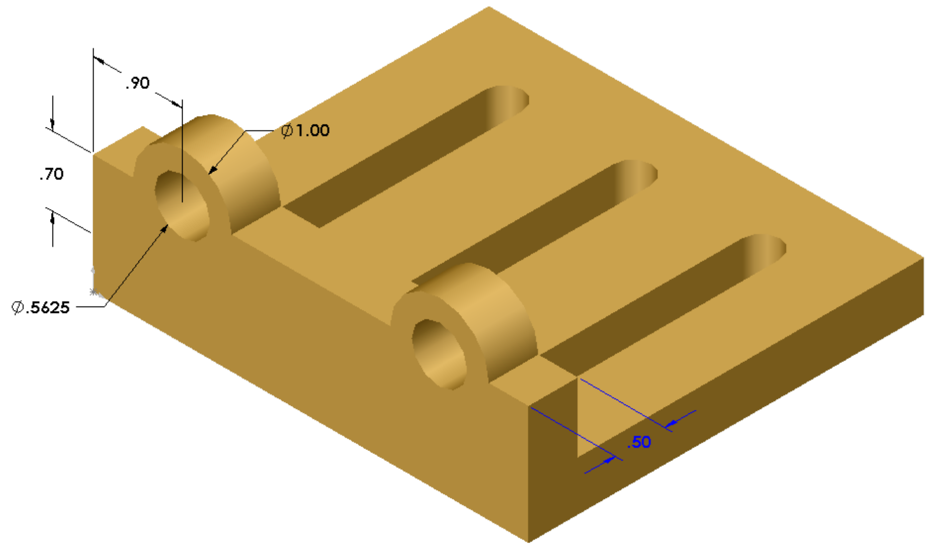
- Use the Rectangular Pattern command to create two more slots on the figure.
- The slots should have a spacing of 1.30” along the long distance of the block.



Exercise 2

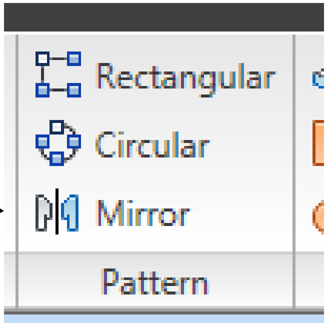
Creation of the Walls(dimensions are typical)

New Sketch on 4.40 side
.70 to top of base



Exercise 2

Creation of the Walls. Using Mirror



- In addition to creating a pattern, a feature can be mirrored around a given plane on the part,

- Select Mirror feature command.
- Select a face or plane for the object to mirror around.
- Select the features you want to mirror.

- To be able to mirror an object the user must have a plane or face to select to mirror around.

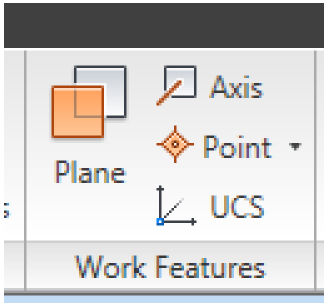
Work Planes

Some of the more common methods of creating work planes are:

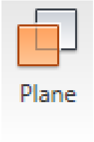
1. Offset
2. Angle
3. Tangent
4. Through line endpoint, Perpendicular
5. Mid-plane between two faces

Work Planes

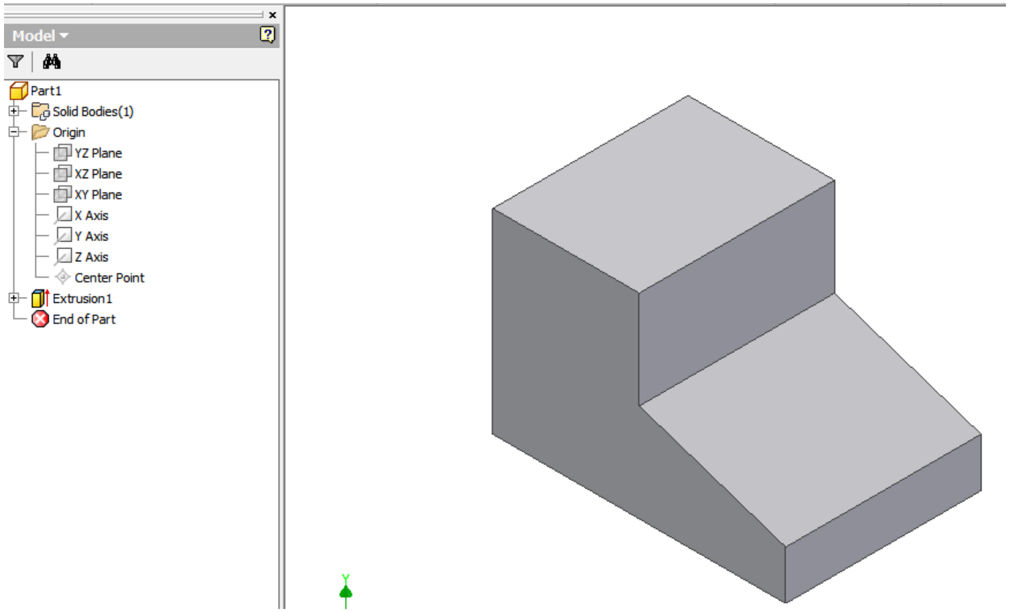
- As we know, to create objects, a sketch is required and a work plane is required to create a sketch.
- A work plane can be used when:
  - A part face is not available as a sketch plane for sketching new features.
  - An intermediate position is required to define other work planes (for example, at an angle to a face at an offset distance).
- An offset work plane can be created from one of the standard yz, yz and xy planes.



Offset Work Plane

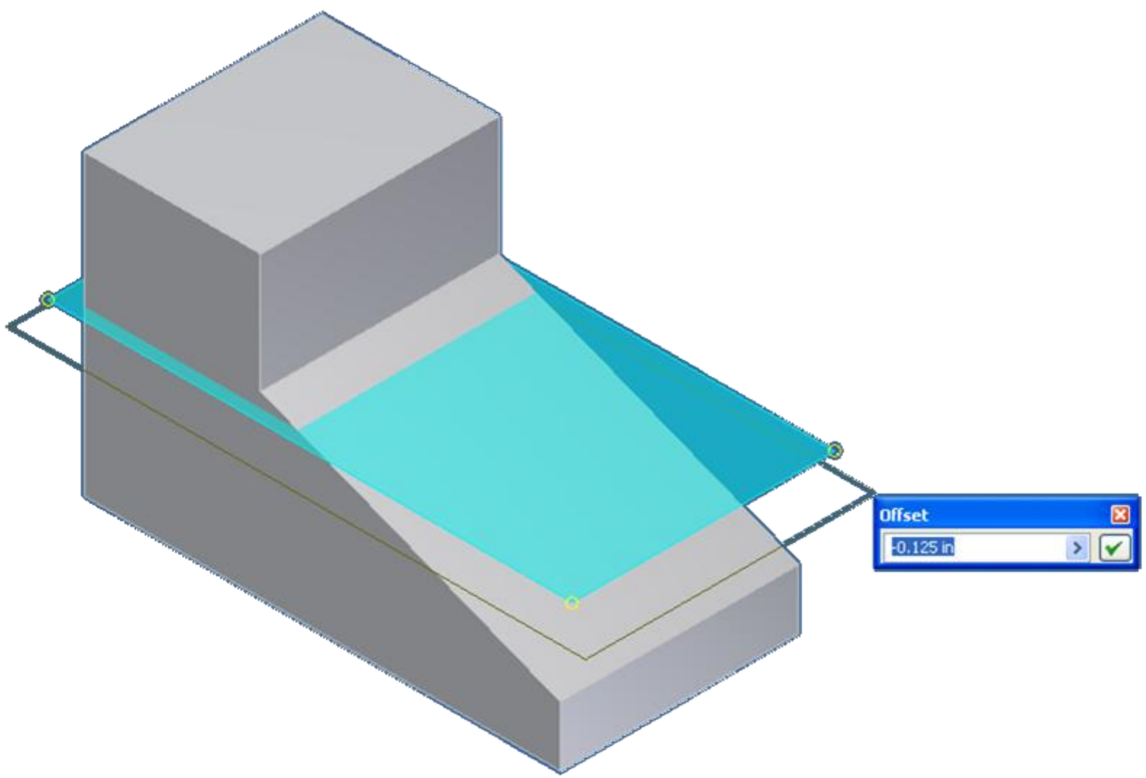
1. Object was created but we needed to drill a vertical hole in slanted surface.
2. Click on work plane and 

3. Select the correct plane required (xz in this case).



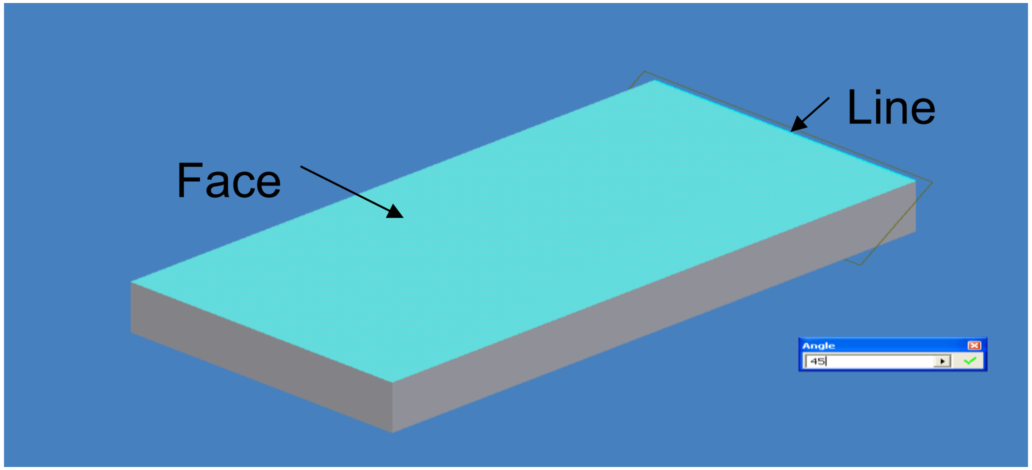
Offset Work Plane

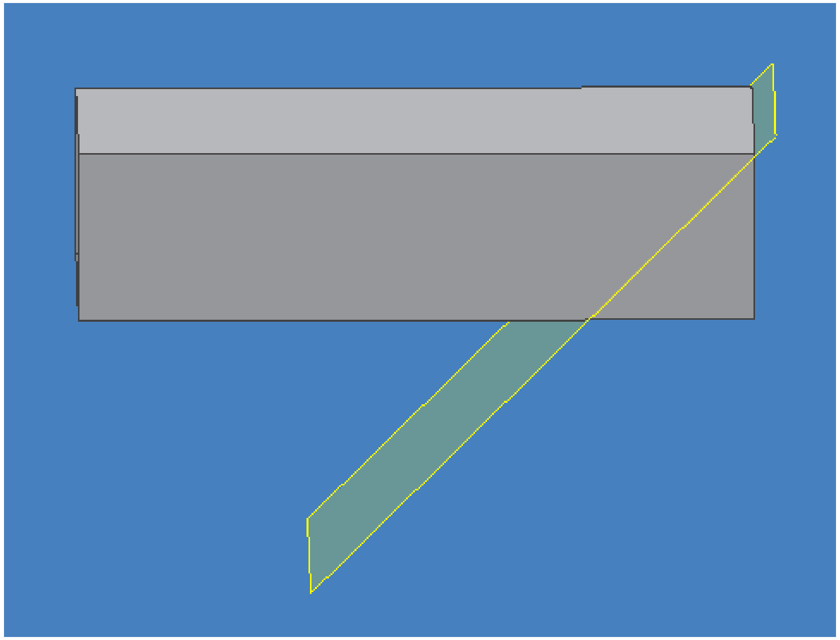
- Select one of the corners
  1. drag to desired location or,
  2. a box will appear and dimension can be inserted.



Angled Work plane

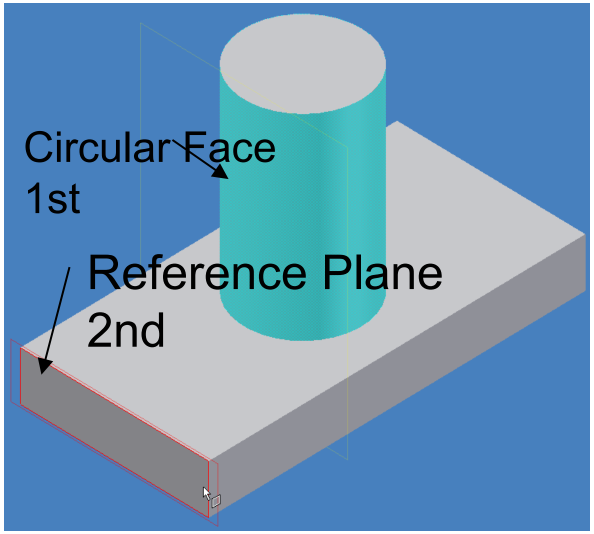
- For an angled plane you need to do the following
  - Select a line or Edge
  - Select a face or plane
  - Fill in the angle you want (45 in this example) in the box.
  - Remember the angle is counterclockwise
  - A new plane will be created

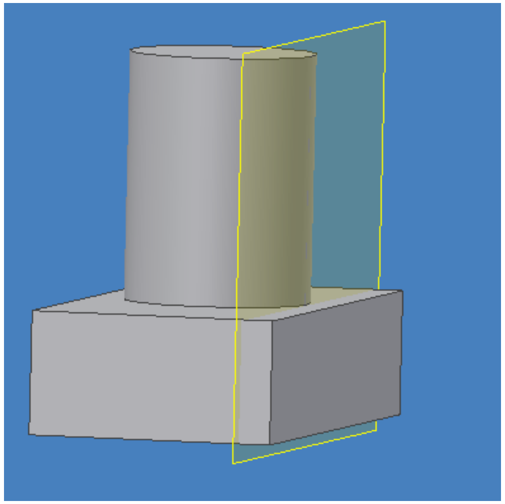




Tangent work plane

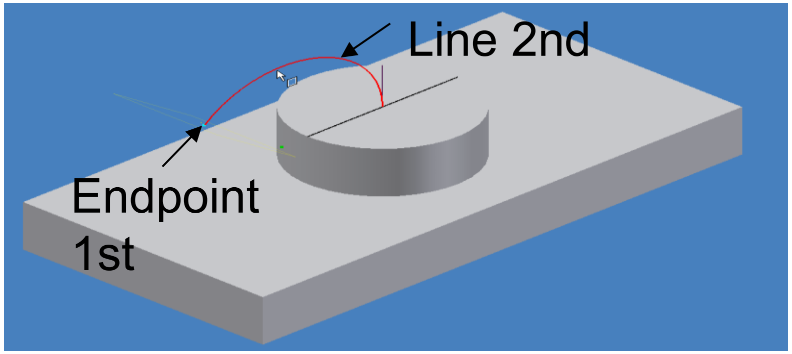
- For a work plane that is tangent to a circular edge or face
  - Select the work plane icon
  - Select the circular face you want your plane tangent to
  - Select the face, or plane that will align your work plane

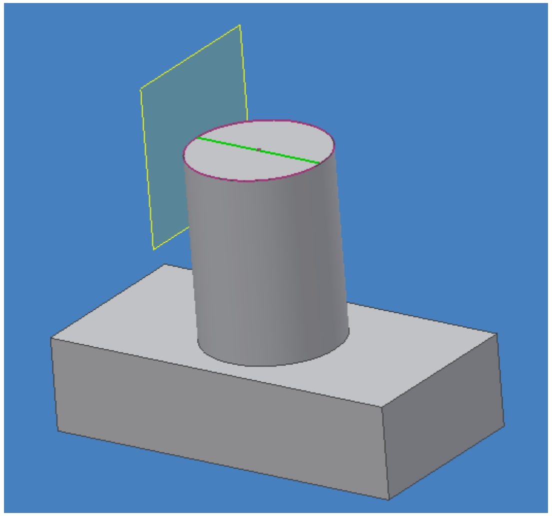


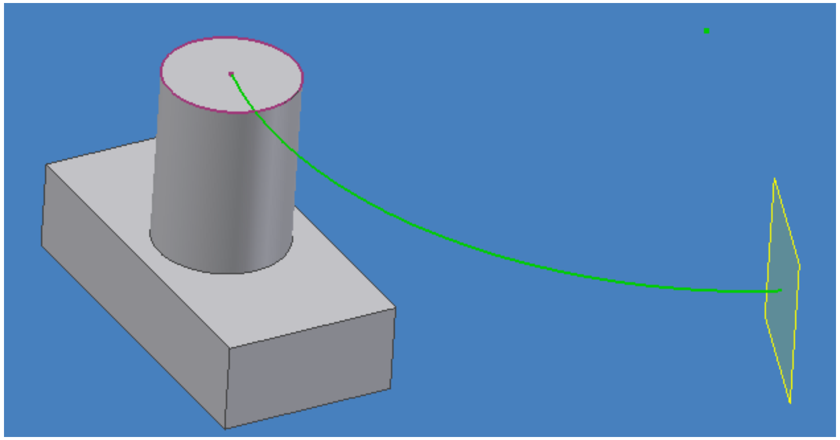


Through line endpoint, Perpendicular

- To create a plane through a particular endpoint and perpendicular to the line you must first have a line sketched or created
  - Select the endpoint of the line
  - Select the line



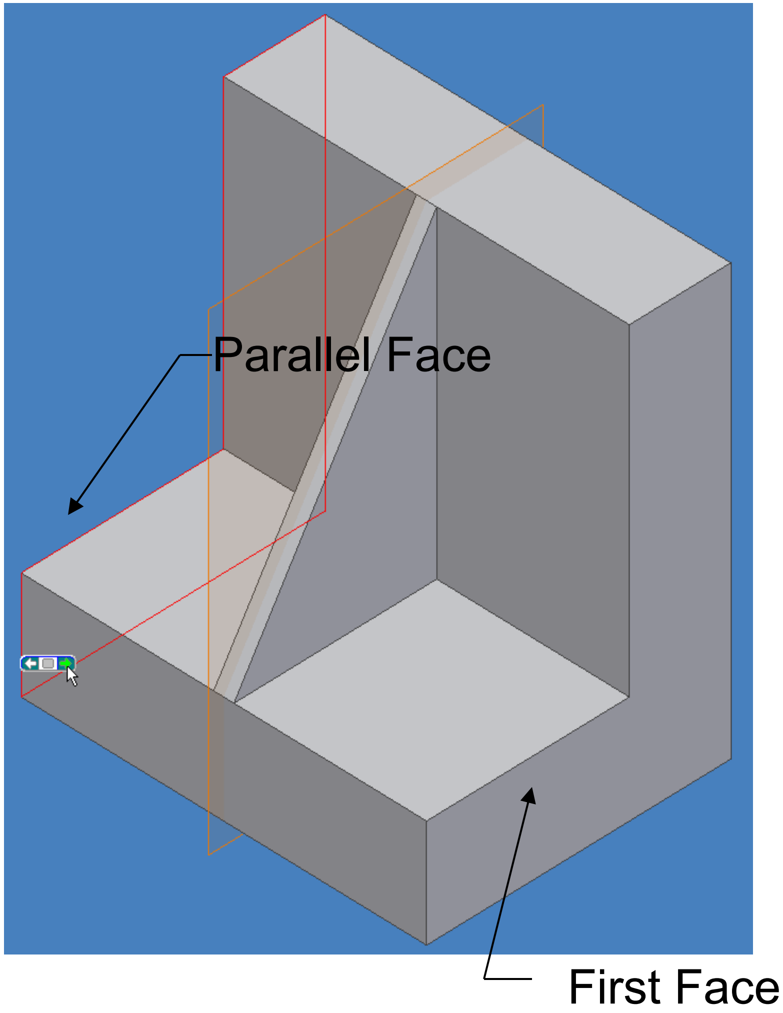




This is particularly useful whenever you are planning on using a sweep or a loft

Mid-plane between two faces

- To place a plane at the midpoint between two parallel faces
  - Select the first face to establish the orientation of the plane
  - Select the parallel face to automatically locate the object at the mid-plane between the two parallel faces

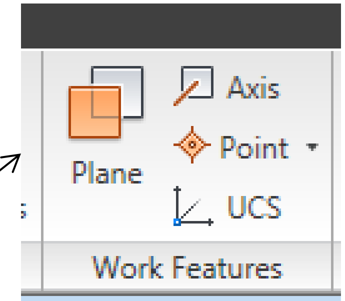


Work planes

- There are two things that are constant with any created work plane.
  1. The work plane's orientation must be defined.
  2. The work plane's location in 3D must be defined.
- With that in mind, there are a lot of ways to create various work planes
  - Angled
  - Edge and face Normal
  - Edge and Tangent
  - Offset
  - Point & Face normal
  - Point & Face Parallel
  - Sketch Geometry
  - Tangent & Face Parallel
  - Edge & Tangent
  - Offset
  - 3 Point
  - 2 Edge
  - Through line endpoint, Perpendicular
- For help, select the work plane icon on the modeling panel and then right click in the modeling window and select "**How To**"
- The help box will show you step by step how to create a new work planes

Exercise 2

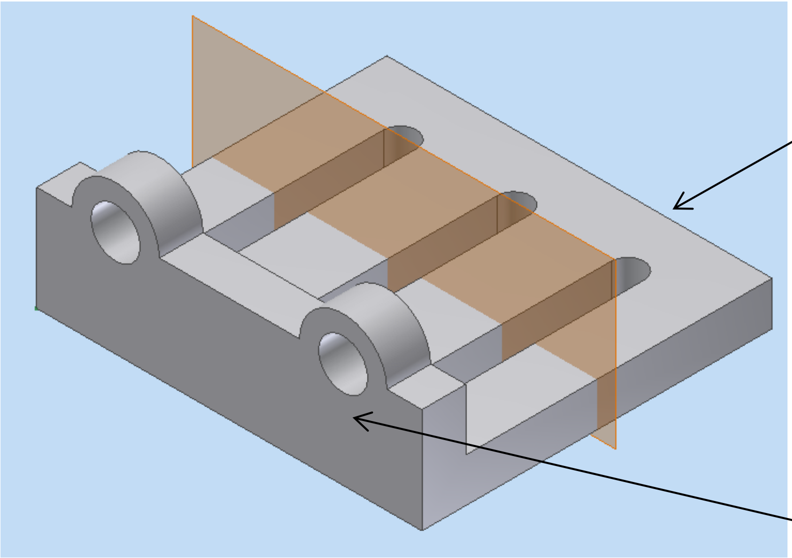
Creation of the Walls. Using Mirror



- To create a construction plane select Plane on the Work Features Panel

- Let's use the mid-plane construction method

Parallel Face

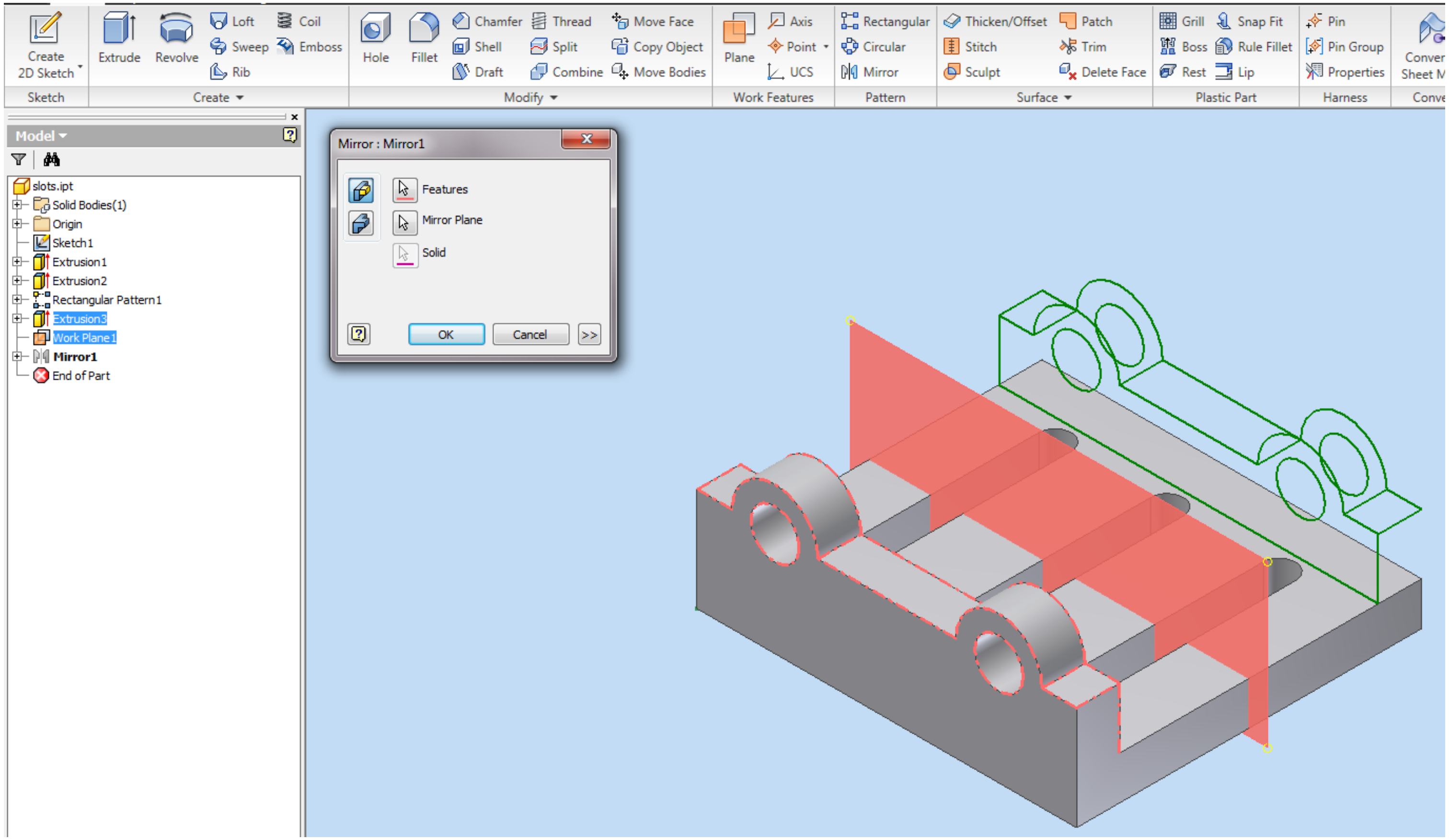


First Face

Exercise 2

Creation of the Walls. Using Mirror

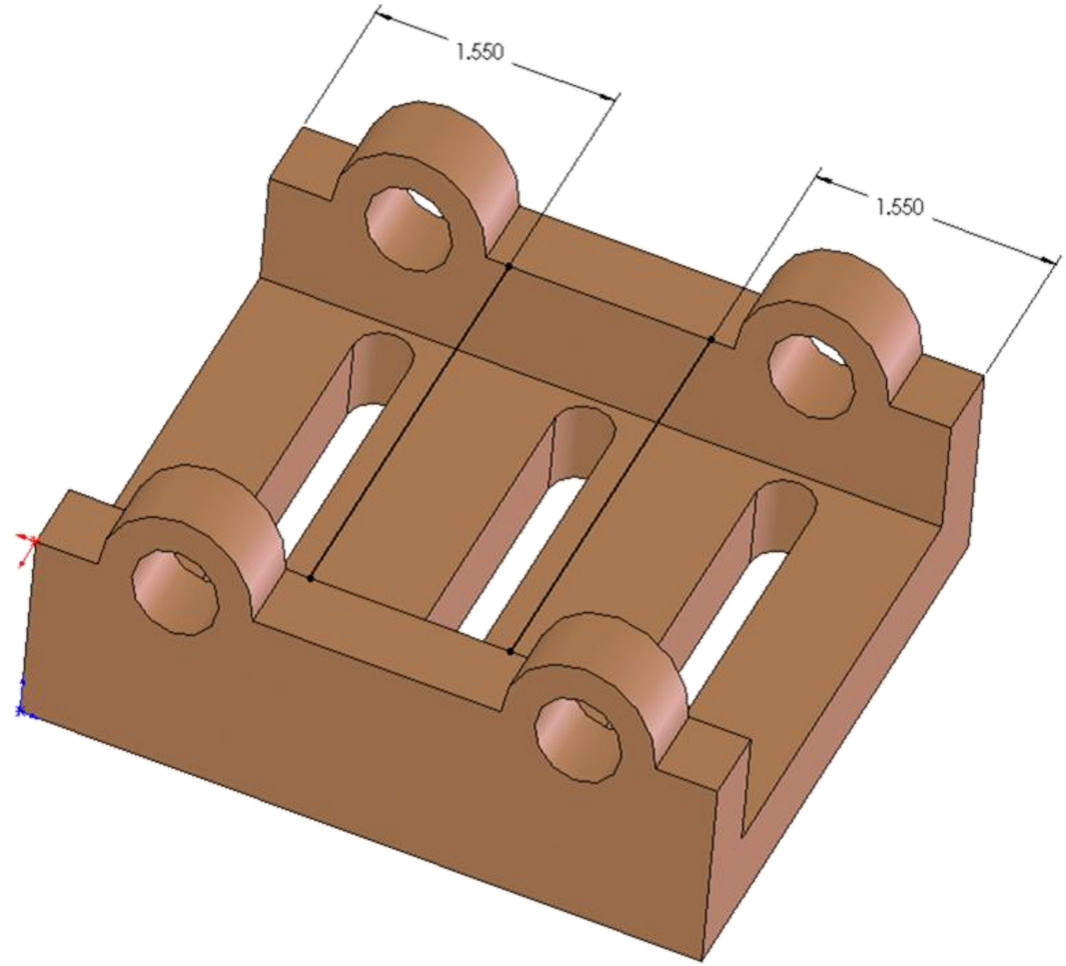
- Once the reference plane is created, you can use the Mirror command



Exercise 2
Adding Ribs

To change the visibility of planes and sketches
Right click on them in the browser(design tree)
And click off visibility

- To complete the part, add two ribs across the uprights.
- Create a sketch on the top surface of the uprights
- Draw two lines and dimension them as shown

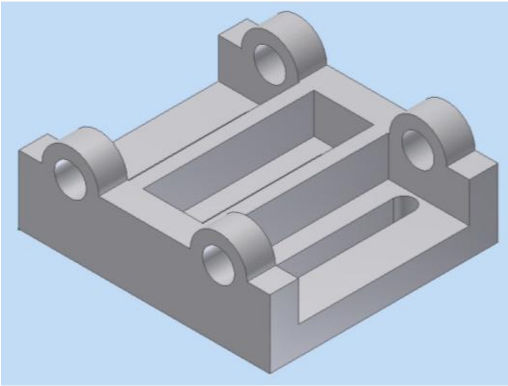


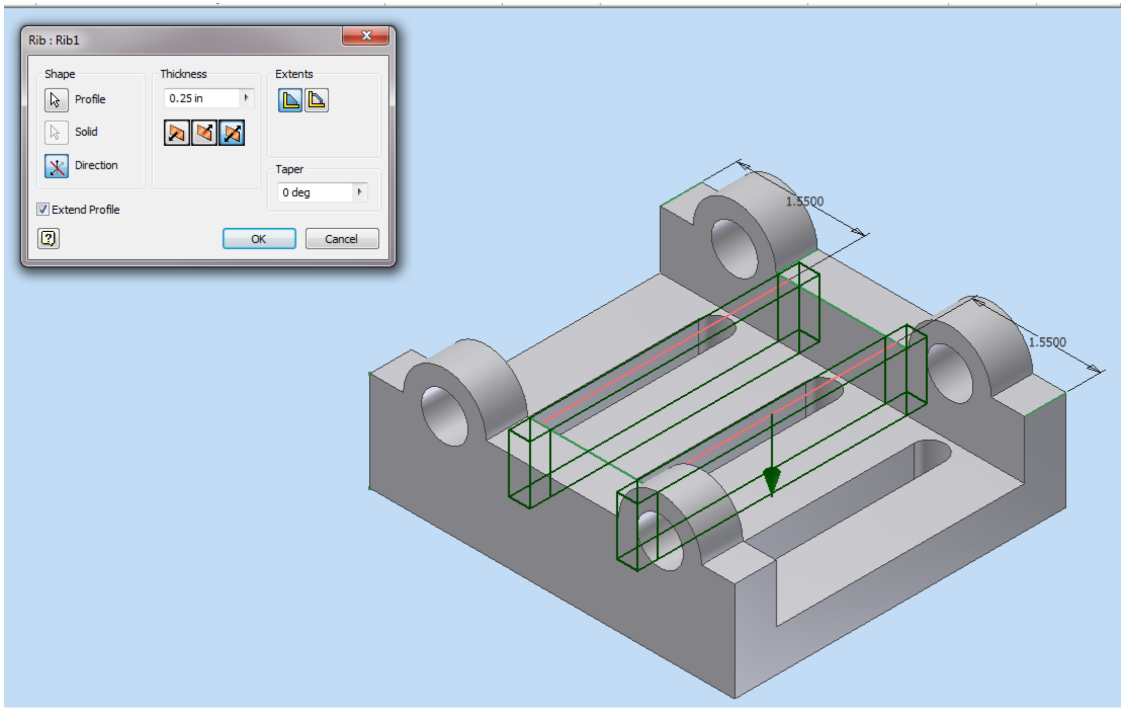
Exercise 2

Adding Ribs

- To complete the ribs, use the rib tool.
- Set the Rib thickness to .25 inches.
- Select the profile

  (two lines)
- Select the direction
  (arrow on screen)
- Select the mid-plane option (material on both sides of line)
- Select OK to complete the Ribs.





Homework Assignment 8.2

- Create a solid model of the Base on page 238 of the textbook.
- Print out an isometric view in a good orientation
- Due at the beginning of the next class period.

References

- Ken Youssefi, “Introduction to Solid Modeling”
- Texas A & M, “Design Intent and Modeling Tools”
- Paul Kurowski, ‘Computer Aided Design (CAD)”